

GRADE 6 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 7) • Forward 2 (Grade 8)
82 question sets • 2465 questions

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Write a ratio that describes the relationship between two quantities

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Reading counts from a table or graph	3.MD.B.3
BACKWARD 2	Parts and totals	2.OA.A.1
BACKWARD 3	Fraction of a whole	3.NF.A.1
FORWARD 1	Ratios and unit rates with fractional quantities	7.RP.A.1
FORWARD 2	The unit rate as the slope of a graph	8.EE.B.5

SET 1 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A fruit bowl has 5 pears and 8 plums.

Write the ratio of pears to plums.

Ratio: _____

A choir has 14 sopranos and 9 altos.

Write the ratio of altos to sopranos.

Ratio: _____

The table shows the pets owned by students in a class.
Write the ratio of dogs to fish.

Pet	Dogs	Cats	Fish
Students	11	6	4

Ratio: _____

A parking lot has 13 cars and 7 trucks. There are no other vehicles.

Write the ratio of trucks to all of the vehicles.

Ratio: _____

A bag has 20 marbles. 12 are green and the rest are yellow.

Write the ratio of yellow marbles to green marbles.

Ratio: _____

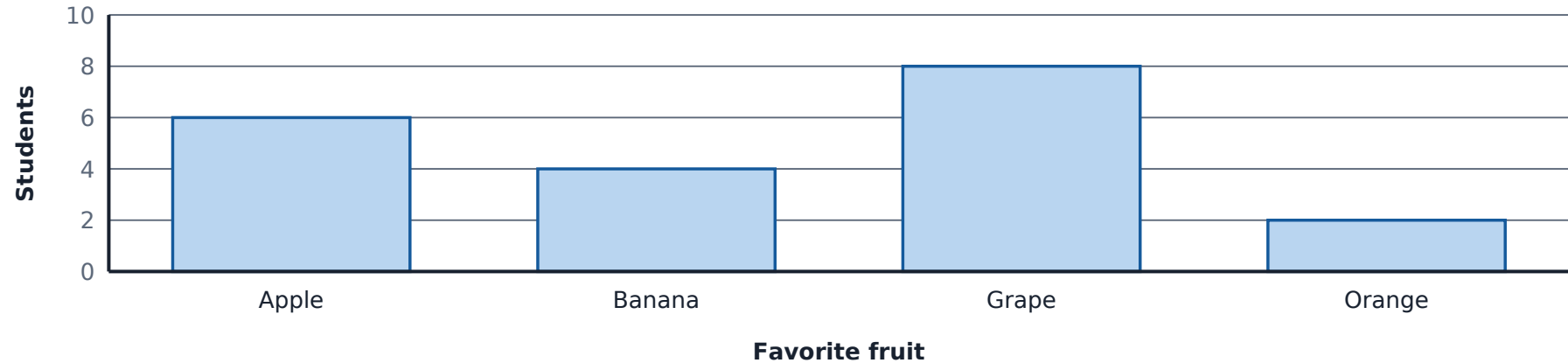
SET 1 • 6.RP.A.1

BACKWARD 1

Reading counts from a table or graph • Grade 3 • Measurement & Data •
3.MD.B.3

The graph shows the favorite fruits of a class.

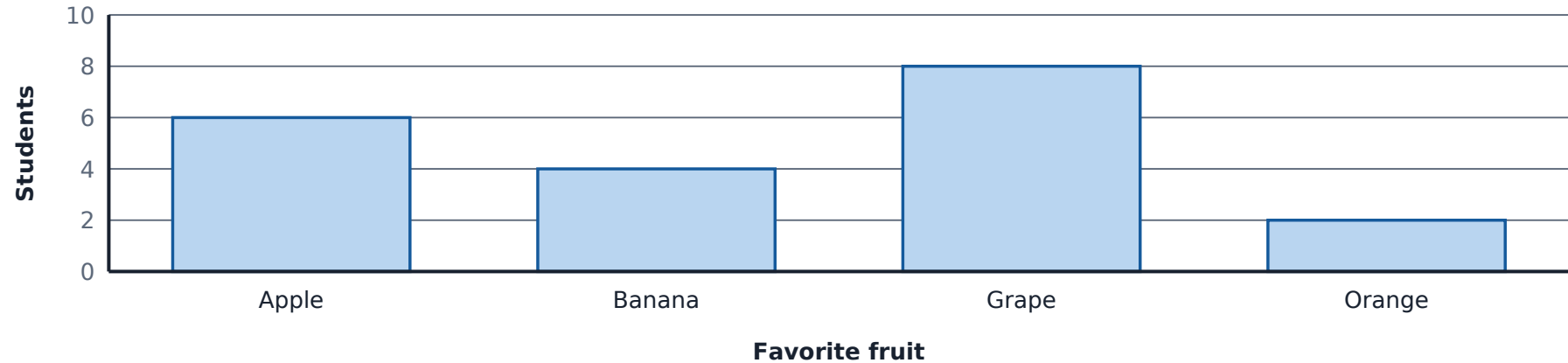
How many students chose grape?



Students: _____

The graph shows the favorite fruits of a class.

How many students chose banana?



Students: _____

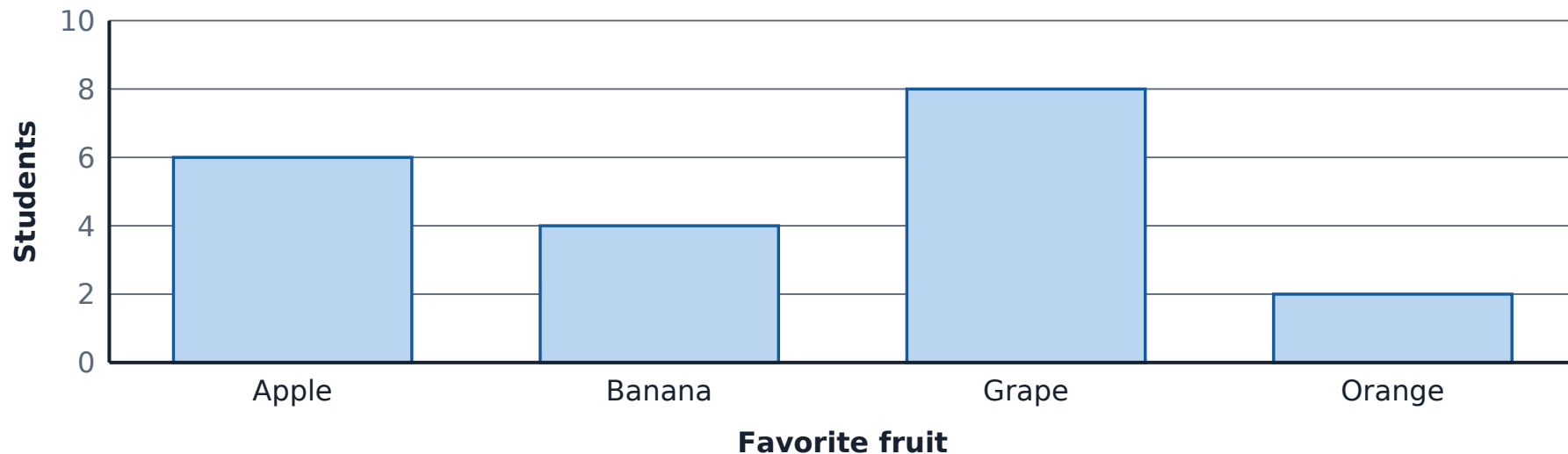
The table shows the pets owned by students.

How many students own cats?

Pet	Dogs	Cats	Fish
Students	11	6	4

Students: _____

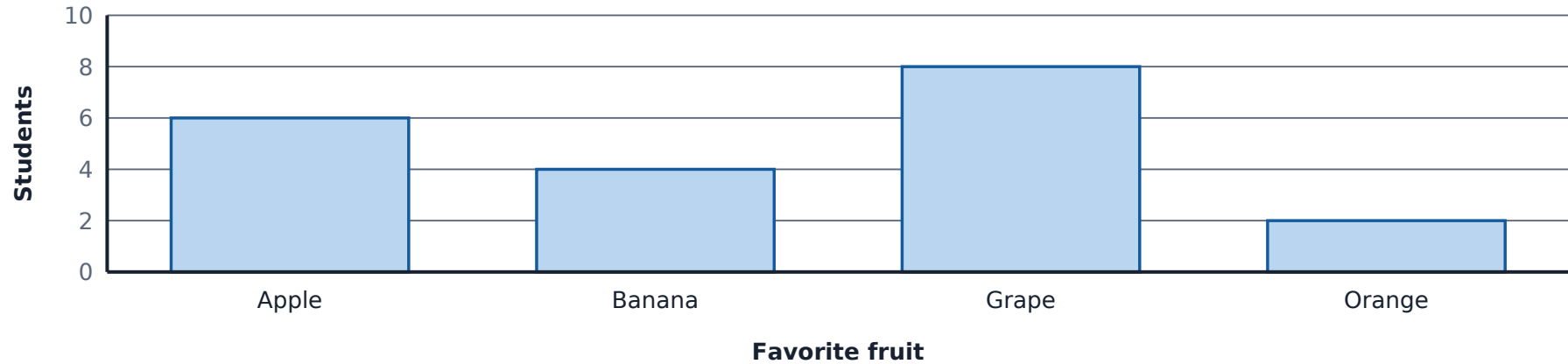
Which fruit was chosen by 2 students?



- A.** Grape **B.** Orange **C.** Banana **D.** Apple

True or false?

6 students chose apple.



A. True

B. False

SET 1 • 6.RP.A.1

BACKWARD 2

Parts and totals • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A team has 9 boys and 6 girls.

How many players are on the team?

Players: _____

A box holds 24 crayons. 15 are broken.

How many crayons are NOT broken?

Crayons: _____

True or false?

A shelf has 7 red books and 12 blue books. There are 19 books in all.

A. True

B. False

A garden has 30 plants. 18 are flowers and the rest are herbs.

How many herbs are there?

A. 12

B. 18

C. 22

D. 48

A bus has 17 adults and 26 children.

How many people are on the bus?

People: _____

SET 1 • 6.RP.A.1

BACKWARD 3

Fraction of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

The bar is divided into equal parts.

What fraction of the bar is shaded?



Fraction: _____

A pizza is cut into 6 equal slices. Ava eats 5 slices.
What fraction of the pizza does Ava eat?

Fraction: _____

True or false?

A bar is cut into 4 equal parts and 1 part is shaded. The shaded part is $\frac{1}{4}$ of the bar.

- A. True
- B. False

What fraction of the bar is shaded?



A. $\frac{2}{5}$

B. $\frac{3}{5}$

C. $\frac{2}{3}$

D. $\frac{5}{2}$

A field has 8 equal sections. Corn grows in 3 of the sections.

What fraction of the field has corn?

Fraction: _____

SET 1 • 6.RP.A.1

FORWARD 1

Ratios and unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A dressing uses $\frac{1}{2}$ cup of oil for every $\frac{3}{4}$ cup of vinegar.

How many cups of oil are used per cup of vinegar?

Cups of oil: _____

Mia paints $\frac{2}{3}$ of a fence in $\frac{1}{2}$ hour.

How much of the fence does she paint per hour?

Fence per hour: _____

A garden uses $\frac{3}{4}$ bag of soil for every $\frac{3}{8}$ bag of mulch.

How many bags of soil are used per bag of mulch?

A. $\frac{9}{32}$

B. $\frac{1}{2}$

C. $1\frac{1}{8}$

D. 2

A cyclist rides $2\frac{1}{2}$ miles in $\frac{1}{6}$ hour.

What is the speed in miles per hour?

Speed: _____

A pump moves $\frac{5}{8}$ gallon of water every $\frac{1}{4}$ minute.

How many gallons does it move per minute?

Gallons per minute: _____

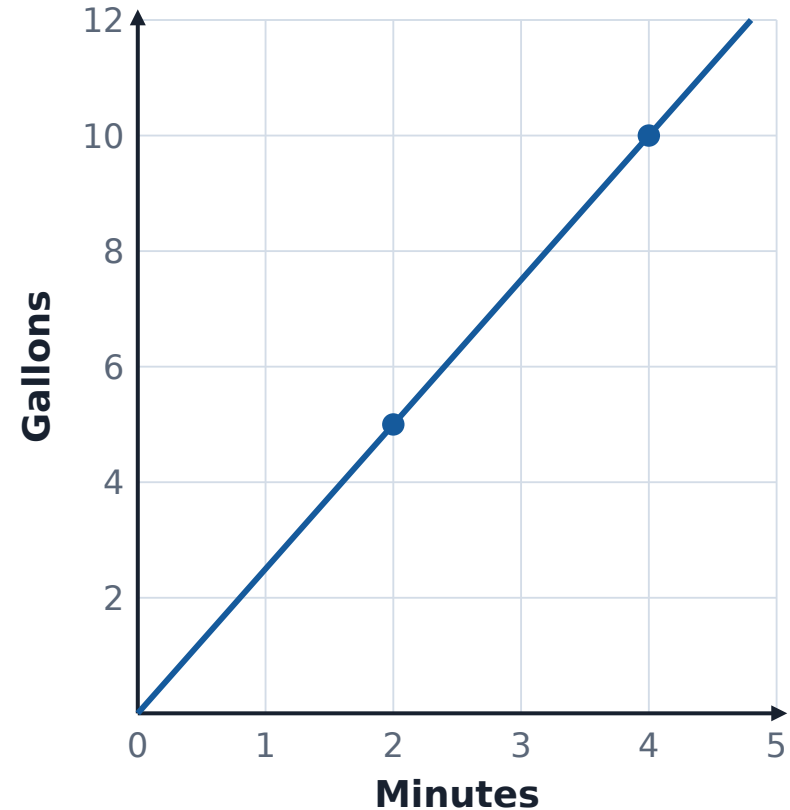
SET 1 • 6.RP.A.1

FORWARD 2

The unit rate as the slope of a graph • Grade 8 • Expressions & Equations •
8.EE.B.5

The graph shows a proportional relationship between minutes and gallons. What is the slope of the line?

Slope: _____



The graph of a proportional relationship passes through $(0, 0)$ and $(6, 9)$.

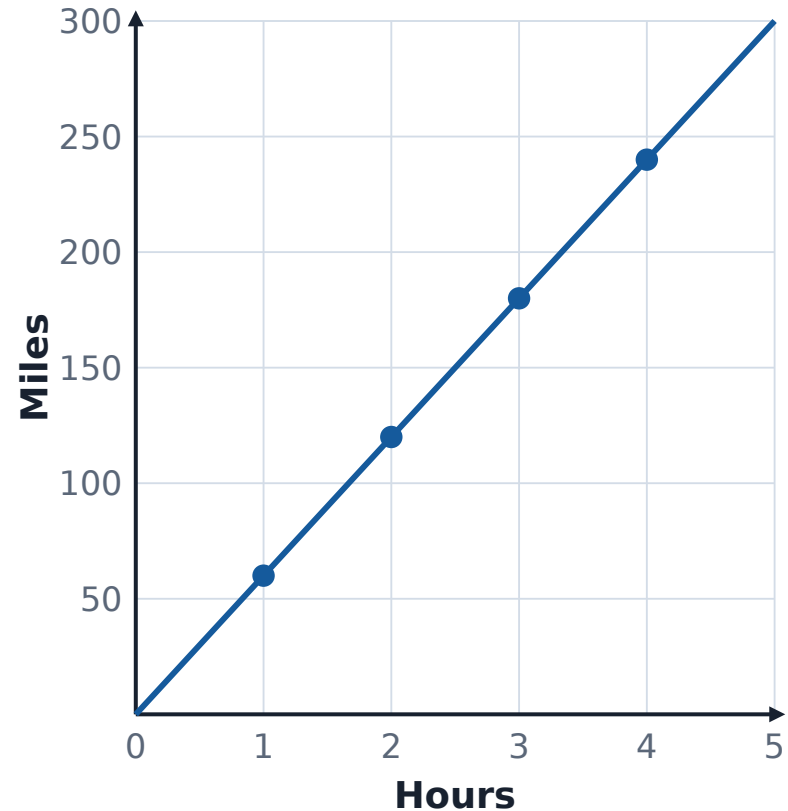
What is the slope?

Slope: _____

The graph shows the distance a car travels.

What is the slope of the line in miles per hour?

Slope: _____



A proportional relationship has a unit rate of 4 dollars per pound.

Which point is on its graph?

A. (3, 12)

B. (4, 1)

C. (3, 7)

D. (12, 3)

True or false?

For $y = 2.5x$, the unit rate is 2.5, and the slope of its graph is also 2.5.

- A. True
- B. False

Interpret ratio language ("for every," "to")

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Equal groups	3.OA.A.1
BACKWARD 2	Multiplicative comparison statements	4.OA.A.1
FORWARD 1	Constant of proportionality from a description	7.RP.A.2.b
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 2 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A recipe uses 5 cups of flour for every 2 cups of sugar.

Which statement describes the same ratio?

- A.** For every 7 cups of sugar, there are 5 cups of flour.
- B.** The ratio of sugar to flour is 2 : 5.
- C.** For every 2 cups of flour, there are 5 cups of sugar.
- D.** The ratio of flour to sugar is 2 : 5.

At an aquarium, the ratio of sharks to rays is 4 : 9.

Which statement is true?

- A.** For every 4 rays, there are 9 sharks.
- B.** For every 9 sharks, there are 4 rays.
- C.** For every 13 sharks, there are 4 rays.
- D.** For every 9 rays, there are 4 sharks.

The ratio of red beads to blue beads is 3 : 7.

Which statement is true?

- A.** For every 10 blue beads, there are 3 red beads.
- B.** For every 7 red beads, there are 3 blue beads.
- C.** For every 3 red beads, there are 7 blue beads.
- D.** For every 3 blue beads, there are 7 red beads.

At a camp, there are 2 counselors for every 9 campers.
What is the ratio of campers to counselors?

- A.** 2 : 9
- B.** 11 : 9
- C.** 9 : 2
- D.** 2 : 11

In a class vote, there were 4 votes for pizza for every 3 votes for tacos.

Which statement is true?

- A.** The ratio of pizza votes to taco votes is 3 : 4.
- B.** The ratio of pizza votes to all votes is 4 : 3.
- C.** There were 4 taco votes for every 3 pizza votes.
- D.** The ratio of taco votes to pizza votes is 3 : 4.

SET 2 • 6.RP.A.1

BACKWARD 1

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

4×6 can mean 4 groups with 6 in each group.

A. True

B. False

There are 3 bags with 5 apples in each bag.

Which expression shows the total?

A. 3×5

B. $5 - 3$

C. $3 + 5$

D. $5 \div 3$

There are 7 tables with 4 chairs at each table.
How many chairs are there?

Chairs: _____

True or false?

"2 wheels for every bike" means each bike has 2 wheels.

A. True

B. False

What does 6×2 mean?

- A.** 6 groups of 2
- B.** 62
- C.** 6 more than 2
- D.** 6 minus 2

SET 2 • 6.RP.A.1

BACKWARD 2

Multiplicative comparison statements • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

True or false?

24 is 3 times as many as 8.

A. True

B. False

Which equation shows "42 is 6 times as many as 7"?

A. $42 = 6 + 7$

B. $42 = 6 \times 7$

C. $42 - 7 = 6$

D. $7 = 6 \times 42$

True or false?

18 is 6 times as many as 3.

A. True

B. False

Fill in the blank.

45 is ____ times as many as 9.

Blank: _____

True or false?

20 is 4 times as many as 16.

A. True

B. False

SET 2 • 6.RP.A.1

FORWARD 1

Constant of proportionality from a description • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A candle burns 2 inches every 3 hours at a constant rate.

What is the constant of proportionality in inches per hour?

$k =$ _____

For every 5 laps, Jo swims 250 meters.

What is the constant of proportionality in meters per lap?

$k =$ _____

A machine fills 18 bottles for every 4 minutes.

What is the constant of proportionality in bottles per minute?

A. $\frac{2}{9}$

B. 4.5

C. 14

D. 72

For every 8 cups of water, a recipe uses 3 cups of rice.

What is the constant of proportionality in cups of rice per cup of water?

$k =$ _____

True or false?

For every 6 pounds of apples, the cost is \$9. The constant of proportionality is 1.5 dollars per pound.

- A.** True
- B.** False

SET 2 • 6.RP.A.1

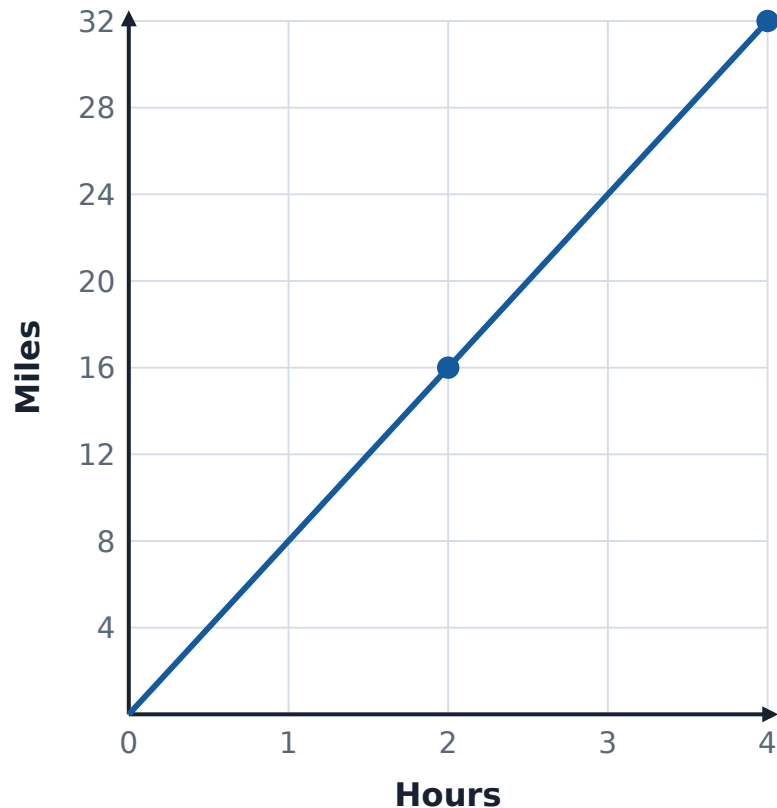
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Cyclist A's distance is shown in the graph. Cyclist B rides according to $d = 7.5t$, where d is miles and t is hours.

Which cyclist is faster?

- A.** Cyclist A
- B.** Cyclist B
- C.** They ride at the same speed.



Tank A fills according to $y = 12x$, where x is minutes and y is gallons. The graph of Tank B is a line through $(0, 0)$ and $(3, 45)$.

Which tank fills faster, and by how many gallons per minute?

Answer: _____

Store A: 2 pounds of apples cost \$3.50.

Store B: the cost is $c = 1.6p$ for p pounds.

Which store charges less per pound?

- A.** Store A
- B.** Store B
- C.** They charge the same.

Printer A prints 3 pages for every 2 seconds. Printer B prints according to $p = 1.4s$.

Which printer is faster?

- A.** Printer A
- B.** Printer B
- C.** They are the same.

Car A travels 150 miles in 3 hours at a constant speed.

Car B travels according to $d = 48t$.

How much faster is Car A, in miles per hour?

Miles per hour: _____

Find and describe a unit rate

MAIN

Grade 6 main question

6.RP.A.2

BACKWARD 1

Whole-number division

4.NBT.B.6

BACKWARD 2

Fraction as division

5.NF.B.3

BACKWARD 3

Meaning of "per" (the amount for each one)

3.OA.A.2

FORWARD 1

Unit rates with fractions

7.RP.A.1

FORWARD 2

Rate of change of a linear function

8.F.B.4

SET 3 • 6.RP.A.2

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A car travels 150 miles in 3 hours at a constant speed.
What is the unit rate in miles per hour?

Unit rate: _____

A 4-pound bag of rice costs \$12.

What is the price per pound?

Price per pound: _____

A punch recipe uses 5 cups of juice for every 8 cups of water.

How many cups of juice are used for each cup of water?

Cups of juice: _____

There are 2 teachers for every 30 students on a field trip.

How many students are there per teacher?

A. 15

B. 28

C. 30

D. 60

A printer prints 90 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

SET 3 • 6.RP.A.2

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Find the quotient.

$$150 \div 3$$

Quotient: _____

Which is equal to $96 \div 4$?

A. 22

B. 23

C. 24

D. 26

True or false?

$$84 \div 6 = 14$$

A. True

B. False

Find the quotient.

$$212 \div 4$$

Quotient: _____

Find the quotient.

$$135 \div 5$$

Quotient: _____

SET 3 • 6.RP.A.2

BACKWARD 2

Fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $5 \div 8$ as a fraction.

Fraction: _____

True or false?

$\frac{2}{5}$ means the same as $2 \div 5$.

A. True

B. False

Four friends share 3 sandwiches equally.

How much sandwich does each friend get?

Each friend gets: _____

Which division expression is equal to $\frac{7}{8}$?

A. $8 \div 7$

B. $8 - 7$

C. $7 \div 8$

D. 7×8

A 5-yard ribbon is cut into 6 equal pieces.
How long is each piece?

Length: _____

SET 3 • 6.RP.A.2

BACKWARD 3

Meaning of "per" (the amount for each one) • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

12 apples are packed equally into 4 bags.

How many apples are in each bag?

Apples: _____

True or false?

"5 miles per hour" means 5 miles for each 1 hour.

A. True

B. False

Which phrase means the same as "\$2 per pound"?

- A.** \$2 for each pound
- B.** \$2 for all of the pounds
- C.** 2 pounds for each dollar
- D.** \$2 more than a pound

A farmer puts 24 eggs into cartons, 6 eggs per carton.
How many cartons does the farmer fill?

Cartons: _____

Jen reads 40 pages in 5 days. She reads the same number of pages each day.

How many pages does she read per day?

Pages: _____

SET 3 • 6.RP.A.2

FORWARD 1

Unit rates with fractions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.1

Kim walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

What is her speed in miles per hour?

Speed: _____

A painter paints $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour.

At this rate, how many walls can the painter paint per hour?

Walls per hour: _____

A snail crawls $\frac{5}{6}$ foot in $\frac{1}{3}$ minute.

How many feet does it crawl per minute?

Feet per minute: _____

A machine fills $\frac{3}{4}$ of a tank in $\frac{1}{2}$ hour.

How many tanks does it fill per hour?

A. $\frac{1}{4}$

B. $\frac{3}{8}$

C. $1\frac{1}{4}$

D. $1\frac{1}{2}$

Leo mows $\frac{7}{8}$ of an acre in $\frac{1}{2}$ hour.

How many acres does he mow per hour?

Acres per hour: _____

SET 3 • 6.RP.A.2

FORWARD 2

Rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What is the rate of change?

x	0	1	2	3
y	5	8	11	14

Rate of change: _____

A plumber charges a \$40 fee plus \$65 per hour.

Write a function for the cost C after h hours.

$C =$ _____

A line passes through $(2, 7)$ and $(6, 19)$.

What is the rate of change?

Rate of change: _____

A tank drains at a constant rate.

What is the rate of change in gallons per minute?

Minutes	0	5	10
Gallons	200	170	140

Rate of change: _____

A taxi ride costs \$3 plus \$2.50 per mile.

Which function gives the cost y for a ride of x miles?

A. $y = 3x + 2.5$

B. $y = 2.5x - 3$

C. $y = 2.5x + 3$

D. $y = 5.5x$

Find missing values in tables of equivalent ratios

MAIN Grade 6 main question

BACKWARD 1 Equivalent fractions

BACKWARD 2 Two related number patterns

BACKWARD 3 Multiples of a number

FORWARD 1 Constant of proportionality from a table

FORWARD 2 Rate of change and initial value from a table

6.RP.A.3.a

4.NF.A.1

5.OA.B.3

4.OA.B.4

7.RP.A.2.b

8.F.B.4

SET 4 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

The table shows equivalent ratios.

Find the missing value.

Cups of juice	2	4	6	10
Cups of water	3	6	9	?

Missing value: _____

Paint is mixed in the same ratio in every column.

Find the missing value.

Blue (cups)	3	6	?	15
Yellow (cups)	2	4	8	10

Missing value: _____

A swimmer keeps the same pace.

Find the missing value.

Laps	3	6	?	15
Minutes	8	16	24	40

Missing value: _____

The table shows equivalent ratios.

What is the missing value?

Tickets	4	8	12	?
Cost (\$)	10	20	30	45

A. 16

B. 18

C. 20

D. 35

A recipe uses 5 cups of oats for every 2 cups of raisins.
Find the missing values A and B.

Oats (cups)	5	10	A	25
Raisins (cups)	2	B	6	10

A = _____

B = _____

SET 4 • 6.RP.A.3.a

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{2}{3}$ is equivalent to $\frac{8}{12}$.

A. True

B. False

Find the missing number.

$$\frac{3}{4} = \frac{?}{20}$$

Missing number: _____

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{5}{12}$

B. $\frac{10}{12}$

C. $\frac{6}{7}$

D. $\frac{10}{11}$

True or false?

$\frac{4}{5}$ is equivalent to $\frac{6}{7}$.

A. True

B. False

Find the missing number.

$$\frac{2}{5} = \frac{6}{?}$$

Missing number: _____

SET 4 • 6.RP.A.3.a

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A starts at 0 and adds 2.

Pattern B starts at 0 and adds 5.

What is the 5th number in Pattern B?

Number: _____

True or false?

Each number in Pattern B is 2 times the matching number in Pattern A.

Pattern A (add 4)	0	4	8	12
Pattern B (add 8)	0	8	16	24

A. True

B. False

Pattern A starts at 0 and adds 7.

Pattern B starts at 0 and adds 14.

What is the 4th number in Pattern A?

Number: _____

Pattern A: 0, 2, 4, 6, 8

Pattern B: 0, 6, 12, 18, 24

How does each number in Pattern B compare to the matching number in Pattern A?

A. 2 times as large

B. 3 times as large

C. 4 more

D. 6 times as large

Pattern A: 0, 1, 2, 3, ...

Pattern B: 0, 4, 8, 12, ...

What number in Pattern B matches 5 in Pattern A?

Number: _____

SET 4 • 6.RP.A.3.a

BACKWARD 3

Multiples of a number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four multiples of 3.

Multiples: _____

True or false?

40 is a multiple of 8.

A. True

B. False

Which number is a multiple of 5?

A. 32

B. 35

C. 48

D. 52

What is the 6th multiple of 4?

Multiple: _____

True or false?

27 is a multiple of 6.

A. True

B. False

SET 4 • 6.RP.A.3.a

FORWARD 1

Constant of proportionality from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality ($\frac{y}{x}$)?

x	2	5	8
y	7	17.5	28

k = _____

The table shows the cost of cheese.

What is the constant of proportionality in dollars per pound?

Pounds	3	6	9
Cost (\$)	7.50	15.00	22.50

$k =$ _____

The table shows a proportional relationship.
What is the constant of proportionality?

x	4	8	12
y	3	6	9

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. 3

D. 12

The table shows how far a toy car rolls.

What is the constant of proportionality in meters per second?

Seconds	2	3	5
Meters	1.2	1.8	3.0

$k =$ _____

True or false?

The constant of proportionality for the table is 6.

x	1	2	4
y	6	12	24

A. True

B. False

SET 4 • 6.RP.A.3.a

FORWARD 2

Rate of change and initial value from a table • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What are the rate of change and the initial value?

x	0	1	2	3
y	4	7	10	13

Rate of change: _____

Initial value: _____

The table shows the height of a plant.

What is the rate of change in cm per week?

Weeks	0	2	4
Height (cm)	6	9	12

Rate of change: _____

The table shows a linear function.

Which equation matches the table?

x	0	1	2	3
y	5	7	9	11

A. $y = x + 5$

B. $y = 5x + 2$

C. $y = 2x + 5$

D. $y = 2x$

The table shows a linear function.

Write an equation for the function.

x	1	2	3	4
y	9	13	17	21

$y =$ _____

True or false?

The initial value of the function in the table is 20.

x	0	1	2
y	20	17	14

A. True

B. False

Plot pairs of values from a ratio table on the coordinate plane

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Ordered pairs from two related patterns	5.OA.B.3
BACKWARD 2	Axes and ordered pairs	5.G.A.1
BACKWARD 3	Plotting points in the first quadrant	5.G.A.2
BACKWARD 4	Reading an axis marked by 2s, 3s, or 5s	3.MD.B.3
FORWARD 1	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a
FORWARD 2	Graph proportional relationships; interpret the slope	8.EE.B.5

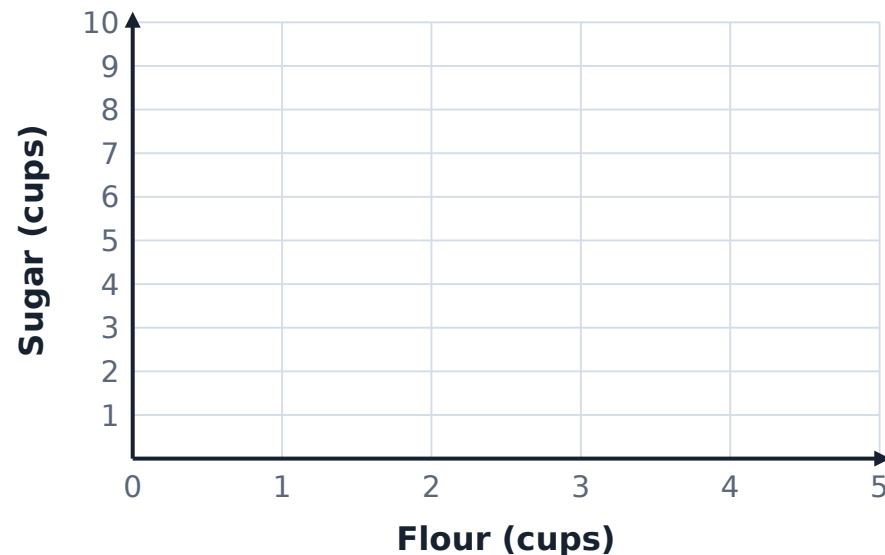
SET 5 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

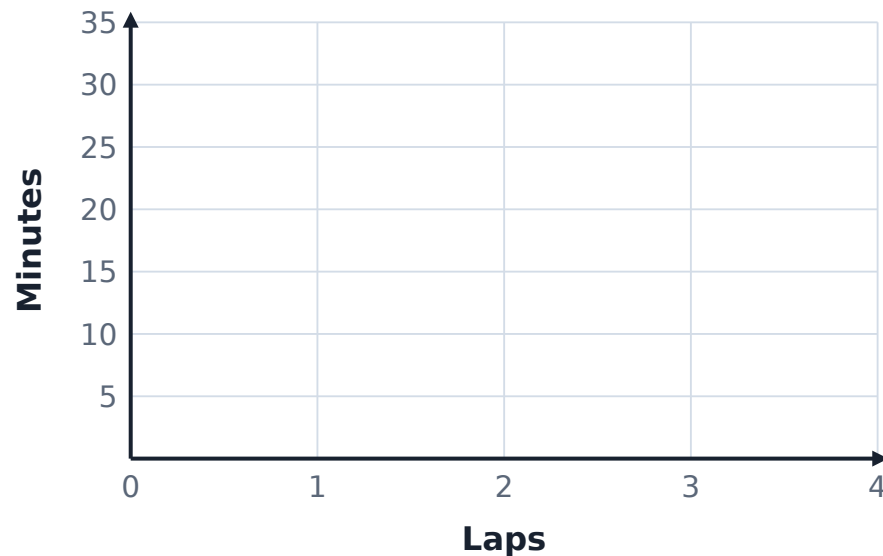
Plot the pairs of values from the table on the coordinate plane.

Flour (cups)	1	2	3	4
Sugar (cups)	2	4	6	8



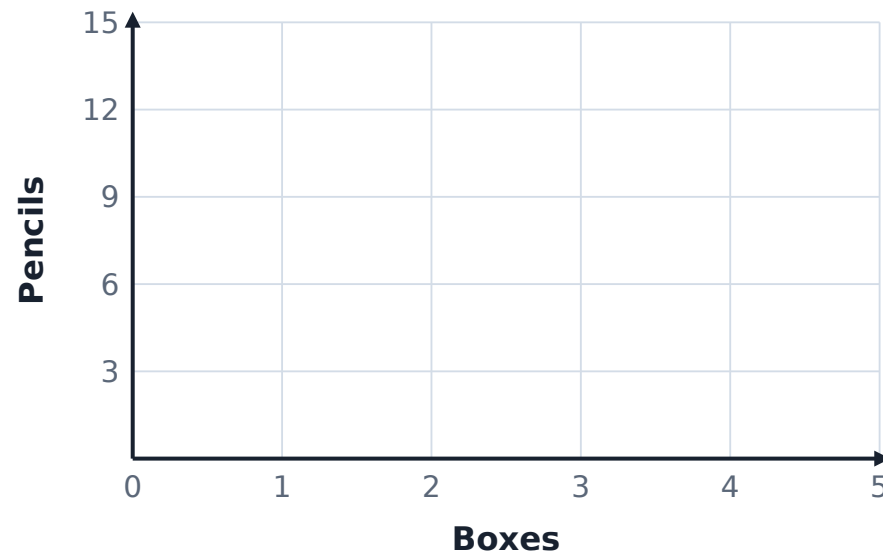
Plot the pairs of values from the table on the coordinate plane.

Laps	1	2	3
Minutes	10	20	30



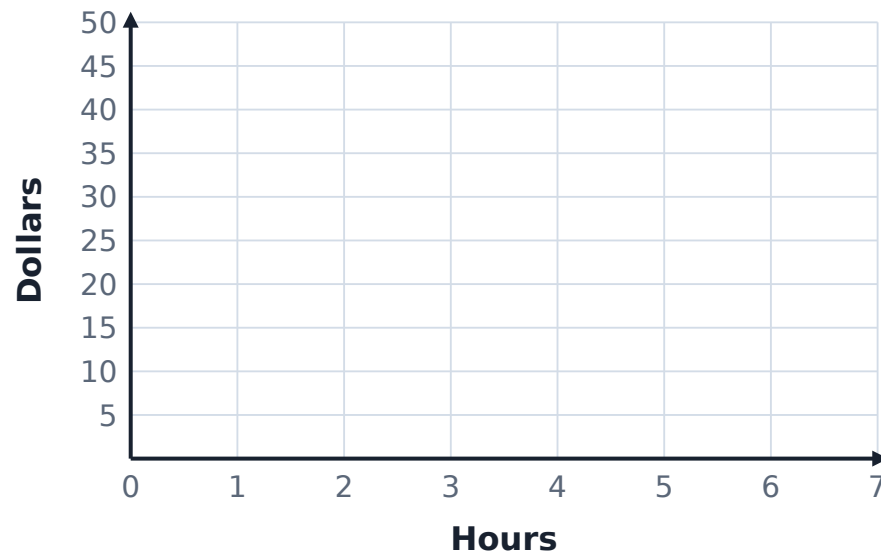
Plot the pairs of values from the table on the coordinate plane.

Boxes	1	2	3	4
Pencils	3	6	9	12



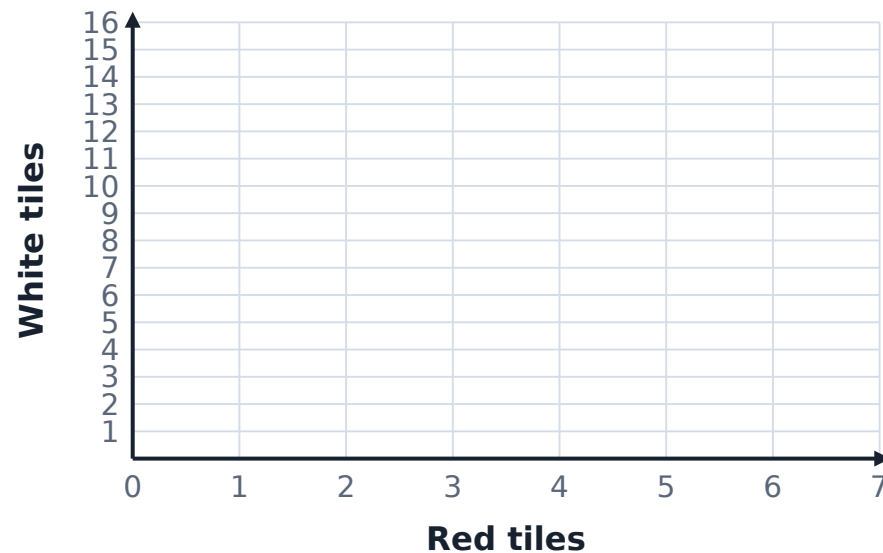
Plot the pairs of values from the table on the coordinate plane.

Hours	2	4	6
Dollars	15	30	45



Plot the pairs of values from the table on the coordinate plane.

Red tiles	2	4	6
White tiles	5	10	15



SET 5 • 6.RP.A.3.a

BACKWARD 1

Ordered pairs from two related patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 3, 6, 9

Write the ordered pair formed by the 3rd numbers of the patterns.

Ordered pair: _____

Which ordered pairs come from the table?

x	1	2	3
y	5	10	15

A. (1, 5), (2, 10), (3, 15)

B. (5, 1), (10, 2), (15, 3)

C. (1, 2), (5, 10), (15, 3)

D. (1, 10), (2, 5), (3, 15)

True or false?

The table gives the ordered pair (4, 12).

x	2	4	6
y	6	12	18

A. True

B. False

Write the ordered pair from the last column of the table.

x	1	2	3	4
y	7	14	21	28

Ordered pair: _____

True or false?

In an ordered pair, the first number comes from the top row of the table.

x	1	2
y	4	8

A. True

B. False

SET 5 • 6.RP.A.3.a

BACKWARD 2

Axes and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot $(3, 7)$, start at $(0, 0)$, move 3 units right, then 7 units up.

A. True

B. False

In the ordered pair (6, 2), which number tells how far to move up?

A. 2

B. 4

C. 6

D. 8

What are the coordinates of the origin?

Origin: _____

True or false?

$(4, 9)$ and $(9, 4)$ name the same point.

A. True

B. False

Which axis is vertical?

A. The x-axis

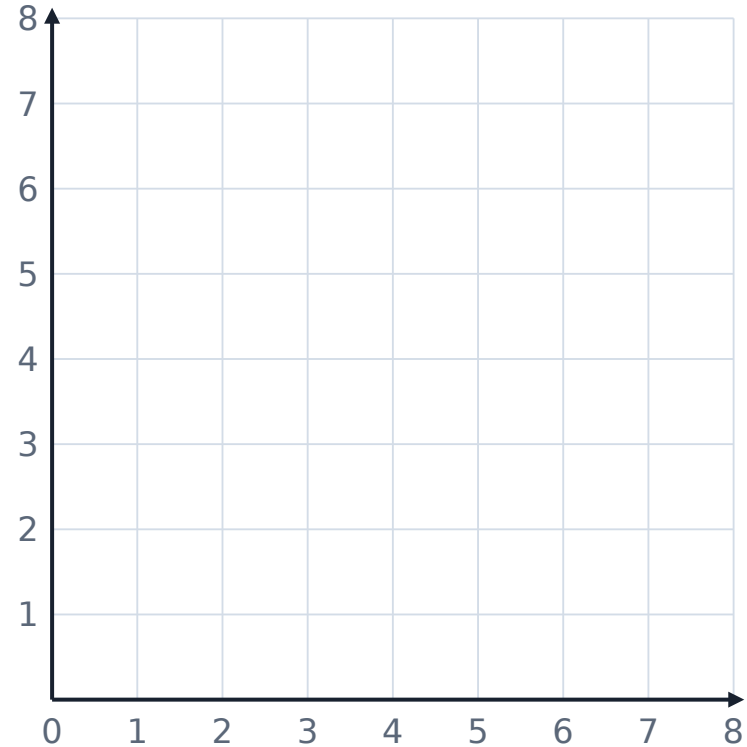
B. The y-axis

SET 5 • 6.RP.A.3.a

BACKWARD 3

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Plot and label point A at (3, 5).

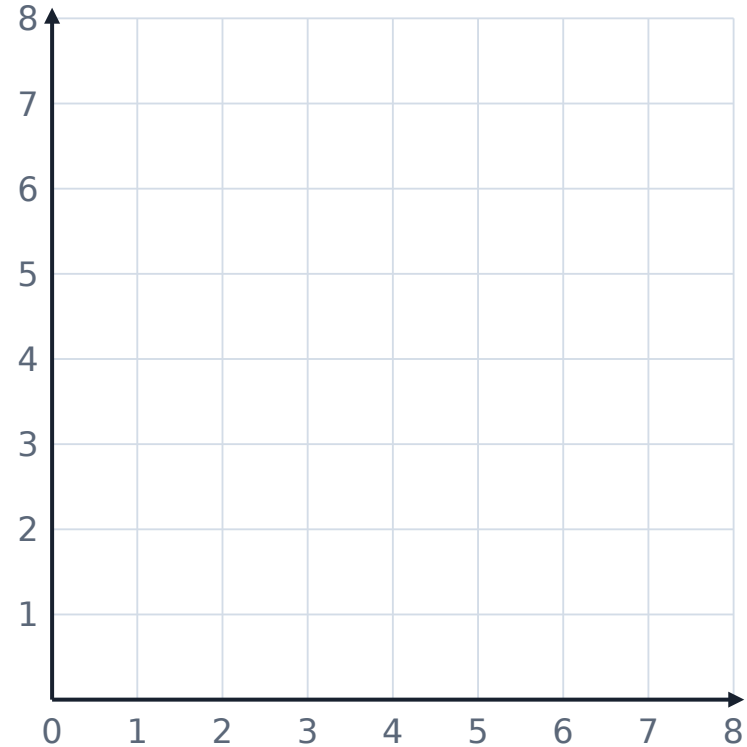


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q1/5

Plot and label point B at (6, 2).

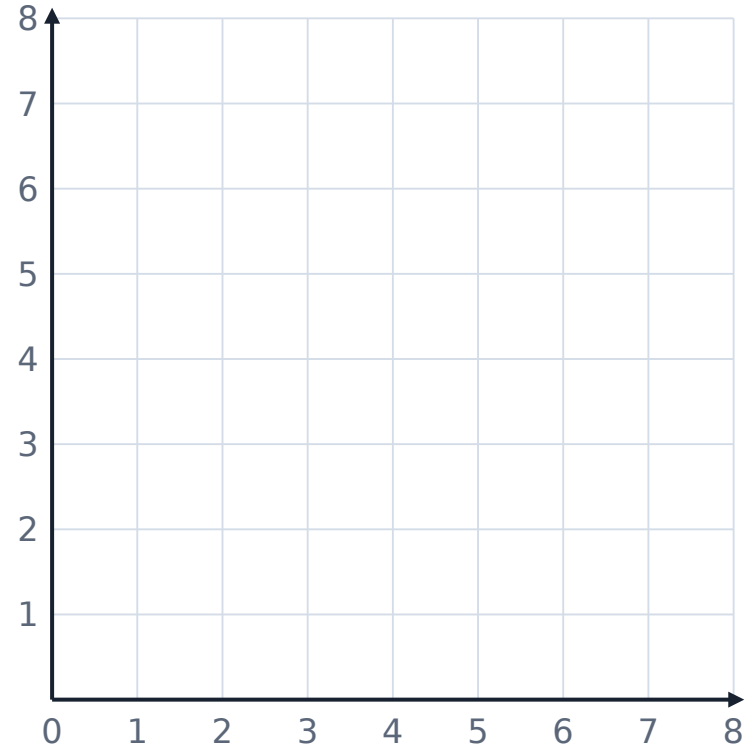


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q2/5

Plot and label point C at (0, 4).



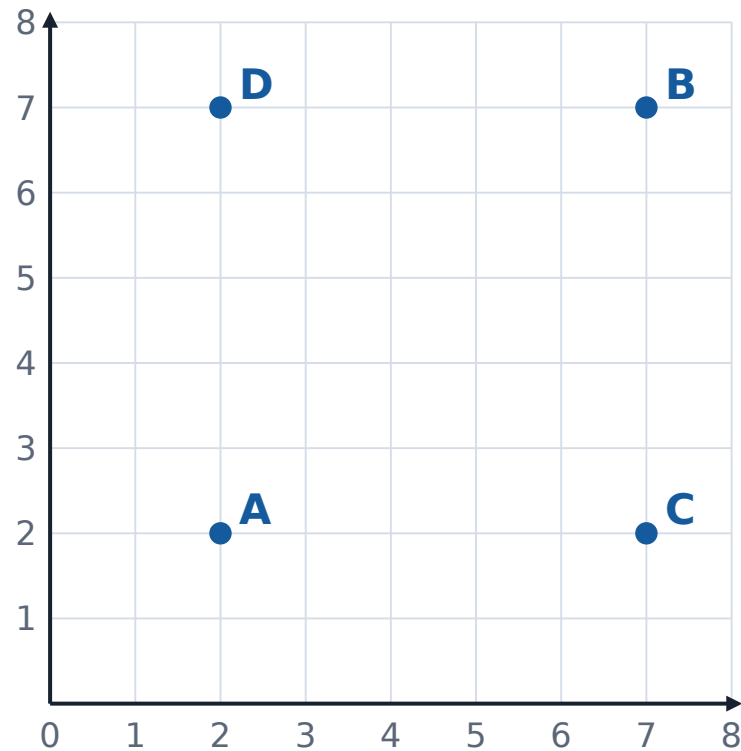
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

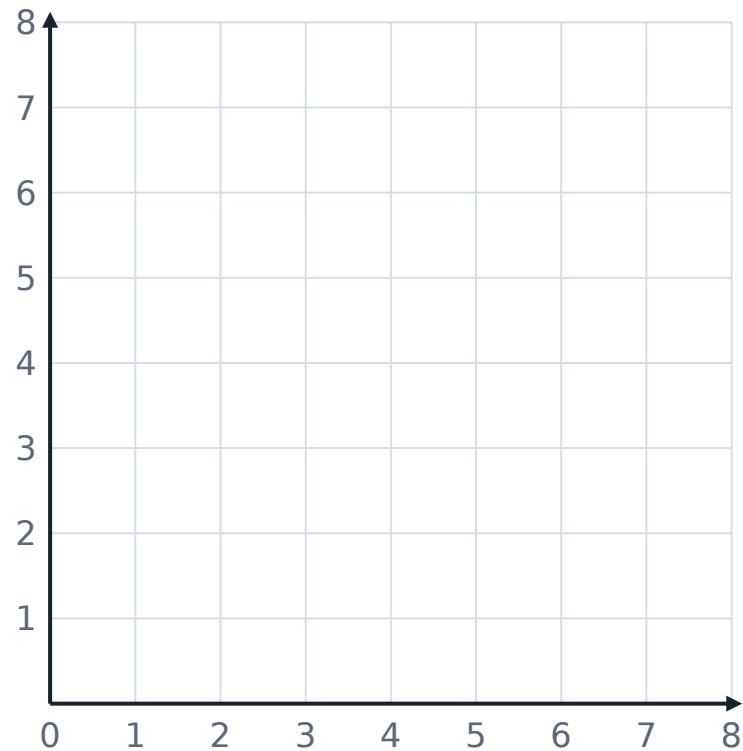
Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q3/5

Which point is at $(2, 7)$?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



Plot and label point D at (5, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 4

Reading an axis marked by 2s, 3s, or 5s • Grade 3 • Measurement & Data •
3.MD.B.3

On the y-axis, the grid lines are numbered 0, 5, 10, 15.
Where would 20 go?

- A.** At the 2nd grid line
- B.** At the 20th grid line
- C.** One grid line above 15
- D.** Halfway between 10 and 15

True or false?

On an axis numbered 0, 3, 6, 9, the value 12 is one grid line above 9.

A. True

B. False

On an axis numbered 0, 2, 4, 6, 8, where does the value 5 go?

Answer: _____

An axis is numbered 0, 5, 10, 15, 20.

How many units does each grid line stand for?

- A.** 1
- B.** 5
- C.** 10
- D.** 20

True or false?

On an axis numbered by 10s, the value 35 is halfway between 30 and 40.

- A.** True
- B.** False

SET 5 • 6.RP.A.3.a

FORWARD 1

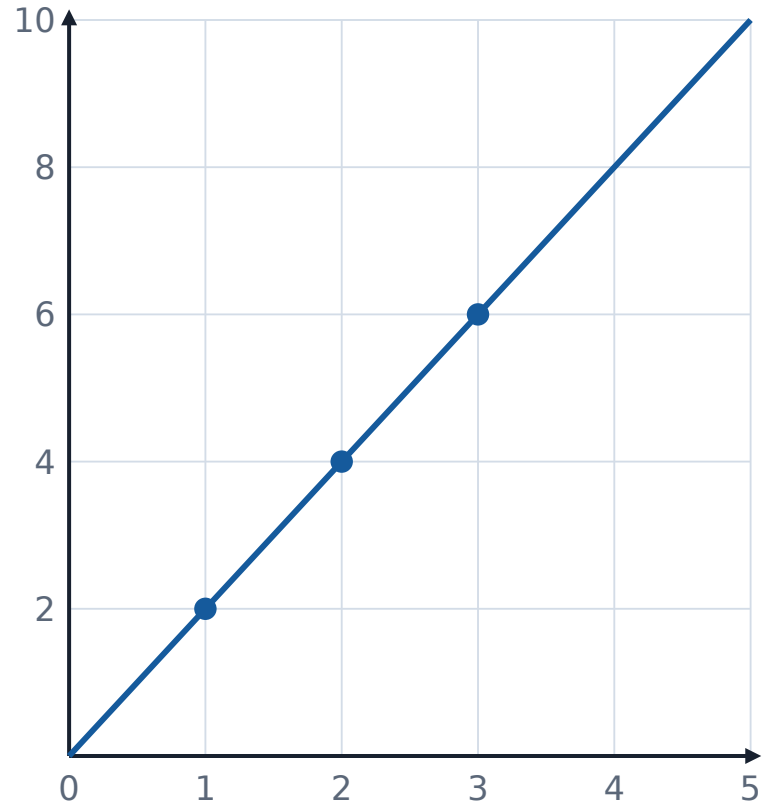
Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship.

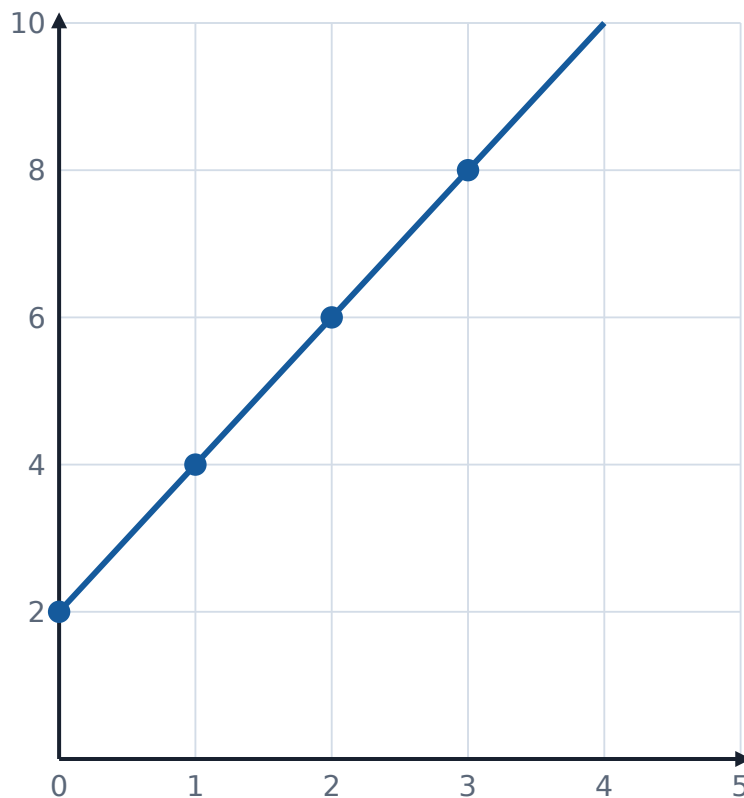
A. True

B. False



Which statement about the graph is true?

- A.** It is proportional because y increases by 2.
- B.** It is not proportional because the line does not pass through $(0, 0)$.
- C.** It is not proportional because the points are not on a line.
- D.** It is proportional because it is a straight line.

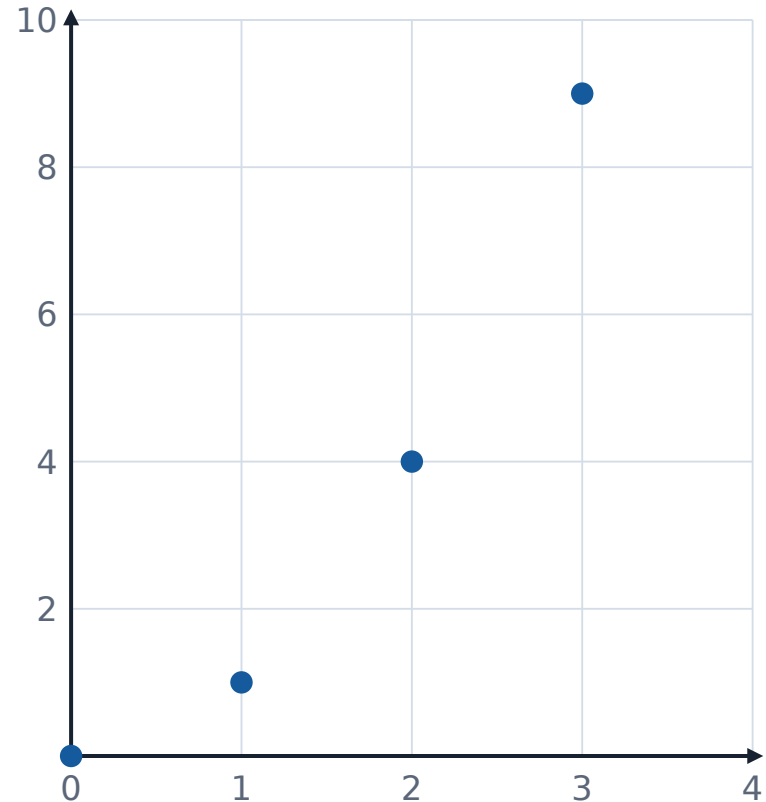


True or false?

The graph shows a proportional relationship.

A. True

B. False



The points $(2, 5)$, $(4, 10)$, and $(6, 15)$ are graphed.

Do they lie on a straight line through the origin? Write yes or no.

Answer: _____

Which set of points lies on a straight line through the origin?

A. $(0, 2), (1, 4), (2, 6)$

B. $(1, 3), (2, 4), (3, 5)$

C. $(2, 3), (4, 6), (6, 9)$

D. $(1, 1), (2, 4), (3, 9)$

SET 5 • 6.RP.A.3.a

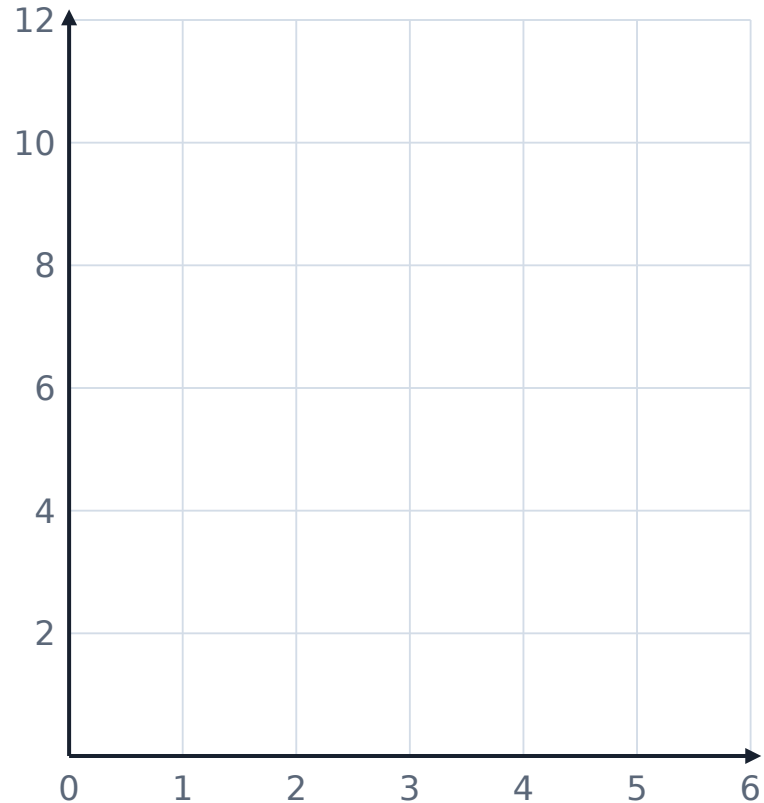
FORWARD 2

Graph proportional relationships; interpret the slope • Grade 8 • Expressions & Equations • 8.EE.B.5

Graph $y = 2x$ on the coordinate plane.

What is the slope of the line?

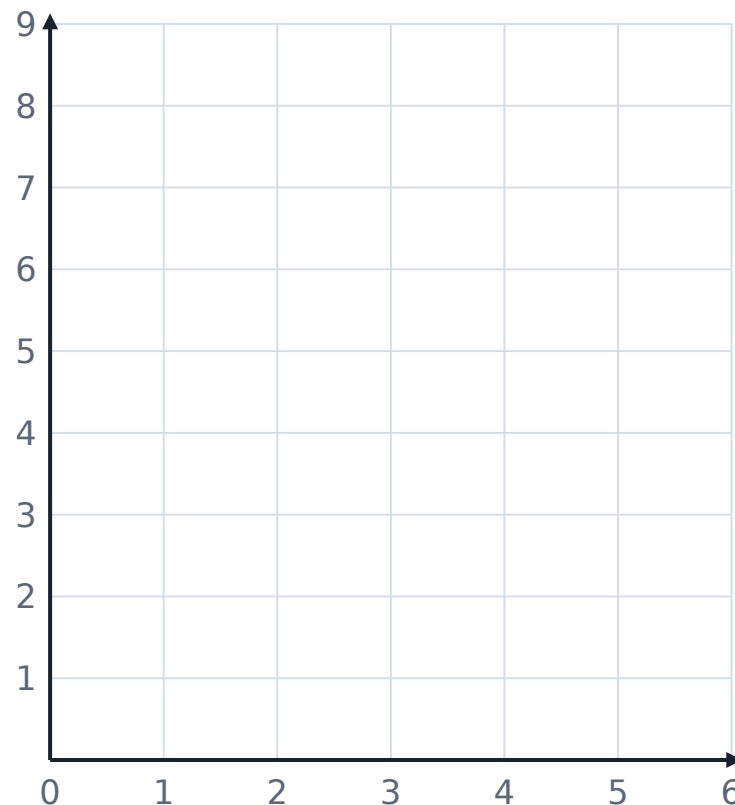
Slope: _____



Graph $y = 1.5x$ on the coordinate plane.

What is the slope of the line?

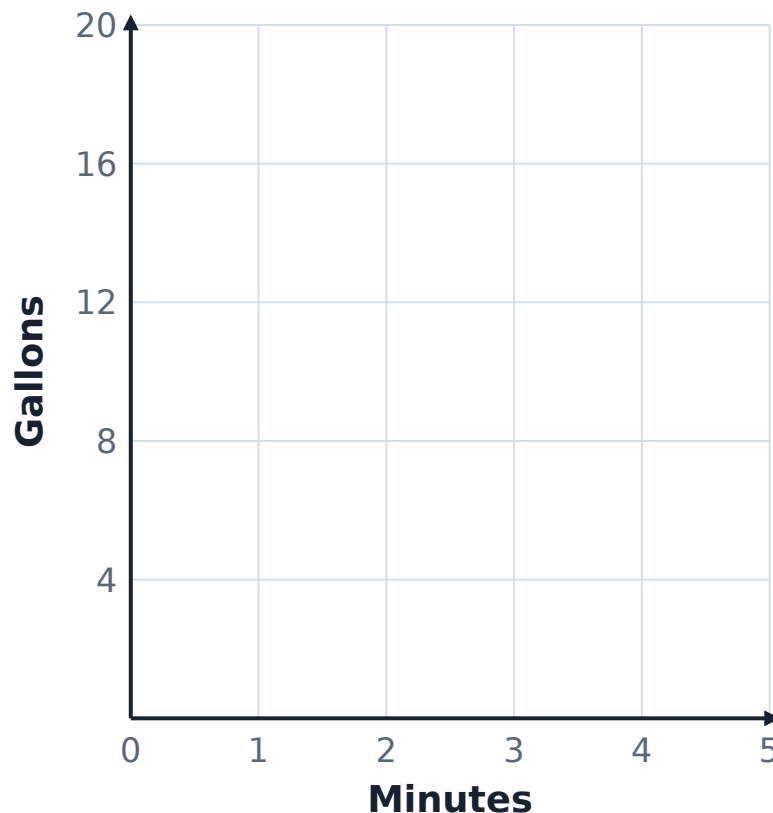
Slope: _____



Water flows at 4 gallons per minute. Graph the gallons y after x minutes.

What is the slope of the line?

Slope: _____



The graph of a proportional relationship passes through (5, 2).

What is the slope?

A. $\frac{2}{5}$

B. 2

C. $\frac{5}{2}$

D. 5

True or false?

The graph of $y = 3x$ passes through $(0, 0)$ and $(2, 6)$, and its slope is 3.

- A. True
- B. False

Use tables of equivalent ratios to compare ratios

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Comparing fractions	4.NF.A.2
BACKWARD 2	Common multiples	4.OA.B.4
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Compare unit rates involving fractions	7.RP.A.1
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 6 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Lemonade A uses 2 cups of mix for every 5 cups of water. Lemonade B uses 3 cups of mix for every 7 cups of water.

Use tables of equivalent ratios. Which lemonade has a stronger mix flavor?

A: Mix	2	4	6
A: Water	5	10	15

B: Mix	3	6	9
B: Water	7	14	21

A. Lemonade A

B. Lemonade B

C. They taste the same.

Class A has 4 boys for every 6 girls. Class B has 6 boys for every 9 girls.

Use tables of equivalent ratios. Which statement is true?

- A.** Class A has a greater ratio of boys to girls.
- B.** Class B has a greater ratio of boys to girls.
- C.** The ratios are equivalent.

Runner A runs 3 laps in 10 minutes. Runner B runs 4 laps in 12 minutes.

Use tables of equivalent ratios. Who runs faster?

- A.** Runner A
- B.** Runner B
- C.** They run at the same speed.

Paint A uses 1 part blue for every 3 parts white. Paint B uses 2 parts blue for every 5 parts white.

Use tables of equivalent ratios. Which paint is a darker blue?

- A.** Paint A
- B.** Paint B
- C.** They are the same color.

Store A sells 5 pens for \$3. Store B sells 8 pens for \$5.

Use tables of equivalent ratios. Which store gives more pens for the same amount of money?

- A.** Store A
- B.** Store B
- C.** They are the same.

SET 6 • 6.RP.A.3.a

BACKWARD 1

Comparing fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which fraction is greater?

A. $\frac{3}{7}$

B. $\frac{3}{5}$

True or false?

$$\frac{5}{8} > \frac{4}{8}$$

A. True

B. False

Which symbol makes the statement true?

$$\frac{2}{3} \text{ — } \frac{3}{4}$$

A. $<$

B. $>$

C. $=$

True or false?

$$\frac{1}{3} = \frac{2}{6}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{1}{2}$

B. $\frac{7}{12}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

SET 6 • 6.RP.A.3.a

BACKWARD 2

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the multiples of 5 and the multiples of 7 up to 40.

Which number is on both lists?

Common multiple: _____

True or false?

30 is a multiple of both 5 and 6.

A. True

B. False

Which number is a multiple of both 3 and 4?

- A.** 9
- B.** 12
- C.** 14
- D.** 16

Name a number that is a multiple of both 6 and 9.

Number: _____

True or false?

20 is a multiple of both 4 and 8.

A. True

B. False

SET 6 • 6.RP.A.3.a

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{5} = \frac{?}{35}$$

Missing number: _____

True or false?

$$\frac{3}{7} = \frac{6}{14}$$

A. True

B. False

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{4}{9}$

B. $\frac{6}{9}$

C. $\frac{6}{8}$

D. $\frac{8}{10}$

Find the missing number.

$$\frac{1}{3} = \frac{5}{?}$$

Missing number: _____

True or false?

$$\frac{3}{10} = \frac{9}{20}$$

A. True

B. False

SET 6 • 6.RP.A.3.a

FORWARD 1

Compare unit rates involving fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{1}{2}$ mile in $\frac{1}{5}$ hour. Ben walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour.

Who walks faster?

- A.** Ana
- B.** Ben
- C.** They walk at the same speed.

Mix A uses $\frac{2}{3}$ cup of sugar per $\frac{1}{2}$ cup of water. Mix B uses $\frac{3}{4}$ cup of sugar per $\frac{2}{3}$ cup of water.

Which mix is sweeter?

- A.** Mix A
- B.** Mix B
- C.** They are the same.

Pump A moves $\frac{3}{8}$ gallon in $\frac{1}{4}$ minute. Pump B moves $\frac{1}{2}$ gallon in $\frac{1}{2}$ minute.

Which pump is faster, and what is its rate in gallons per minute?

Answer: _____

Snail A crawls $\frac{1}{6}$ foot in $\frac{1}{2}$ minute. Snail B crawls $\frac{1}{4}$ foot in $\frac{2}{3}$ minute.

Which snail is faster?

- A.** Snail A
- B.** Snail B
- C.** They are the same.

True or false?

$\frac{1}{2}$ mile in $\frac{1}{4}$ hour is a faster speed than $\frac{3}{4}$ mile in $\frac{1}{2}$ hour.

A. True

B. False

SET 6 • 6.RP.A.3.a

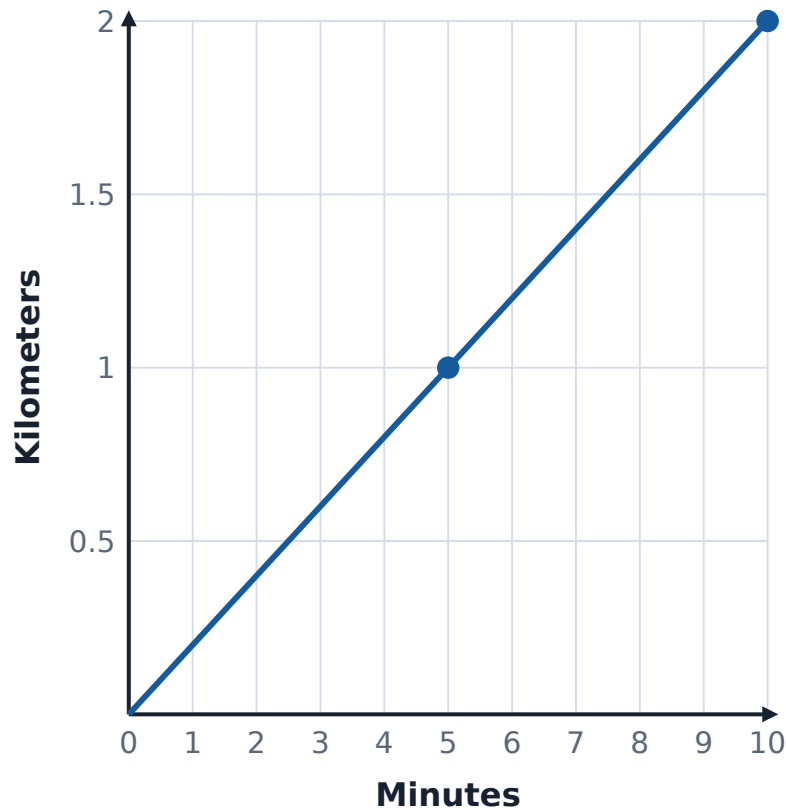
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Runner A's distance is shown in the graph. Runner B runs according to $d = 0.15t$, where d is kilometers and t is minutes.

Which runner is faster?

- A.** Runner A
- B.** Runner B
- C.** They are the same.



Machine A: $y = 45x$ bottles after x minutes. Machine B: the table shows 3 minutes $\rightarrow$ 120 bottles, 6 minutes $\rightarrow$ 240 bottles.

Which machine is faster?

- A.** Machine A
- B.** Machine B
- C.** They are the same.

Recipe A uses flour and sugar in the ratio $y = 1.5x$.

Recipe B's graph passes through $(0, 0)$ and $(4, 7)$.

Which recipe uses more flour y per cup of sugar x ?

Recipe: _____

Two proportional relationships: P is $y = 6x$. Q is a line through $(0, 0)$ and $(2, 13)$.

Which has the greater slope?

- A.** P
- B.** Q
- C.** They are the same.

Plan A: 20 minutes of calls cost \$3.00. Plan B:
 $c = 0.12m$, where m is minutes.

Which plan costs less per minute?

Plan: _____

Solve unit pricing problems

MAIN	Grade 6 main question
BACKWARD 1	Dividing money amounts
BACKWARD 2	Multiplying money amounts
BACKWARD 3	Finding the cost of one when each costs the same
FORWARD 1	Equations for proportional relationships
FORWARD 2	Build a linear function for a pricing situation

6.RP.A.3.b

5.NBT.B.7

5.NBT.B.7

3.OA.A.2

7.RP.A.2.c

8.F.B.4

SET 7 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

4 notebooks cost \$10.

At this rate, how much do 7 notebooks cost?

Cost: _____

3 pounds of apples cost \$5.40.

At this rate, how much do 5 pounds cost?

Cost: _____

6 cans of soup cost \$7.50.

At this rate, how much do 10 cans cost?

Cost: _____

5 movie tickets cost \$42.50.

At this rate, how much do 3 tickets cost?

- A.** \$8.50
- B.** \$25.50
- C.** \$28.33
- D.** \$127.50

8 yards of fabric cost \$36.

At this rate, how much do 3 yards cost?

Cost: _____

SET 7 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$10.00 \div 4$$

Quotient: _____

Find the quotient.

$$\$5.40 \div 3$$

Quotient: _____

$$\$7.50 \div 6 = ?$$

- A.** \$0.80
- B.** \$1.20
- C.** \$1.25
- D.** \$12.50

Find the quotient.

$$\$36 \div 8$$

Quotient: _____

True or false?

$$\$42.50 \div 5 = \$8.50$$

A. True

B. False

SET 7 • 6.RP.A.3.b

BACKWARD 2

Multiplying money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Multiply.

$$\$2.50 \times 7$$

Product: _____

Multiply.

$$\$1.80 \times 5$$

Product: _____

True or false?

$$\$1.25 \times 10 = \$12.50$$

A. True

B. False

$$\$4.50 \times 3 = ?$$

- A.** \$1.50
- B.** \$7.50
- C.** \$12.50
- D.** \$13.50

Multiply.

$$\$8.50 \times 3$$

Product: _____

SET 7 • 6.RP.A.3.b

BACKWARD 3

Finding the cost of one when each costs the same • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

4 toys cost \$12 in all. Each toy costs the same.

How much does 1 toy cost?

Cost: _____

5 pencils cost 30 cents in all. Each pencil costs the same.

How much does 1 pencil cost?

Cost: _____

True or false?

6 stickers cost 18 cents in all. Each costs the same, so 1 sticker costs 3 cents.

A. True

B. False

8 cards cost \$16. Each card costs the same.

How much is 1 card?

- A.** \$2
- B.** \$8
- C.** \$24
- D.** \$128

3 books cost \$27 in all. Each book costs the same.

How much does 1 book cost?

Cost: _____

SET 7 • 6.RP.A.3.b

FORWARD 1

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Apples cost \$2.50 per pound.

Write an equation for the cost c of p pounds of apples.

Equation: _____

The table shows a proportional relationship.

Write an equation for the earnings e after h hours.

Hours (h)	2	3	5
Earnings (e)	36	54	90

Equation: _____

Pens cost \$0.75 each.

Which equation gives the cost c of n pens?

A. $c = \frac{n}{0.75}$

B. $c = n + 0.75$

C. $c = 0.75n$

D. $n = 0.75c$

5 pounds of rice cost \$8.

Write an equation for the cost c of p pounds of rice.

Equation: _____

True or false?

Grapes cost \$3.20 per pound. The equation $c = 3.2p$ gives the cost c of p pounds.

- A. True
- B. False

SET 7 • 6.RP.A.3.b

FORWARD 2

Build a linear function for a pricing situation • Grade 8 • Functions • 8.F.B.4

A store charges \$2.50 per pound of cherries plus a \$4 box fee.

Write a function for the total cost C of p pounds.

$C =$ _____

Tickets cost \$12 each plus a \$5 service fee per order.

Which function gives the cost C of t tickets?

A. $C = 5t + 12$

B. $C = 12t - 5$

C. $C = 12t + 5$

D. $C = 17t$

A delivery costs \$6 plus \$1.75 per mile.

What are the rate of change and the initial value?

Rate of change: _____

Initial value: _____

The table shows the cost of a gym membership.
Write a function for the cost C after m months.

Months (m)	0	1	2	3
Cost (C)	25	55	85	115

$C =$ _____

True or false?

A function for "\$3 per pound plus a \$2 bag fee" is $C = 3p + 2$.

- A. True
- B. False

Solve constant speed problems

MAIN	Grade 6 main question
BACKWARD 1	Whole-number division
BACKWARD 2	Multiplying by a one-digit number
BACKWARD 3	Hours and minutes
FORWARD 1	Speed as a unit rate with fractions
FORWARD 2	Compare speeds from graphs and equations

6.RP.A.3.b

4.NBT.B.6

4.NBT.B.5

4.MD.A.1

7.RP.A.1

8.EE.B.5

SET 8 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

A runner runs 12 kilometers in 60 minutes at a constant speed.

How far does the runner go in 25 minutes?

Distance: _____

A train travels 180 miles in 3 hours at a constant speed.
How far does it travel in 5 hours?

Distance: _____

A cyclist rides 45 miles in 3 hours at a constant speed.
How long does it take to ride 75 miles?

Time: _____

A snail moves 18 cm in 6 minutes at a constant speed.
How far does it move in 20 minutes?

- A.** 3 cm
- B.** 54 cm
- C.** 60 cm
- D.** 120 cm

A car travels 220 miles in 4 hours at a constant speed.
How long does it take to travel 385 miles?

Time: _____

SET 8 • 6.RP.A.3.b

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$180 \div 3$$

Quotient: _____

Divide.

$$220 \div 4$$

Quotient: _____

True or false?

$$385 \div 55 = 7$$

A. True

B. False

$$45 \div 3 = ?$$

- A.** 12
- B.** 14
- C.** 15
- D.** 135

Divide.

$$375 \div 5$$

Quotient: _____

SET 8 • 6.RP.A.3.b

BACKWARD 2

Multiplying by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$60 \times 5$$

Product: _____

Multiply.

$$15 \times 7$$

Product: _____

True or false?

$$55 \times 4 = 220$$

A. True

B. False

$$48 \times 6 = ?$$

- A.** 54
- B.** 248
- C.** 288
- D.** 2,448

Multiply.

$$125 \times 3$$

Product: _____

SET 8 • 6.RP.A.3.b

BACKWARD 3

Hours and minutes • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

True or false?

There are 60 minutes in 1 hour.

A. True

B. False

Complete.

5 hours = ____ minutes

Minutes: _____

Which is longer?

- A.** 1 hour
- B.** 90 minutes
- C.** They are the same.

How many minutes are in $\frac{1}{2}$ hour?

Minutes: _____

SET 8 • 6.RP.A.3.b

FORWARD 1

Speed as a unit rate with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A hiker walks $\frac{3}{4}$ mile in $\frac{3}{8}$ hour.

What is the hiker's speed in miles per hour?

Speed: _____

A turtle crawls $\frac{1}{5}$ mile in $\frac{1}{2}$ hour.

What is its speed in miles per hour?

Speed: _____

A boat travels $2\frac{1}{2}$ miles in $\frac{1}{4}$ hour.

What is its speed in miles per hour?

A. $\frac{5}{8}$

B. $2\frac{3}{4}$

C. 5

D. 10

A skater glides $\frac{2}{3}$ kilometer in $\frac{1}{6}$ hour.

What is the skater's speed in kilometers per hour?

Speed: _____

True or false?

Going $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 5 miles per hour.

A. True

B. False

SET 8 • 6.RP.A.3.b

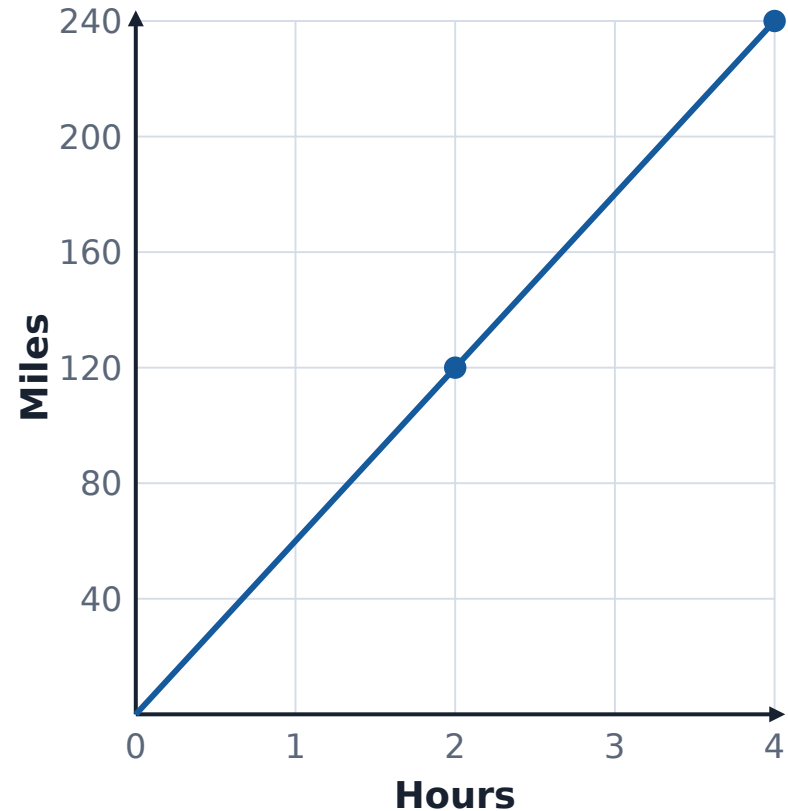
FORWARD 2

Compare speeds from graphs and equations • Grade 8 • Expressions & Equations • 8.EE.B.5

Car A's distance is shown in the graph. Car B travels according to $d = 55t$.

Which car is faster?

- A.** Car A
- B.** Car B
- C.** They are the same.



Train A travels according to $d = 80t$. Train B's graph is a line through $(0, 0)$ and $(3, 210)$.

Which train is faster, and by how many miles per hour?

Answer: _____

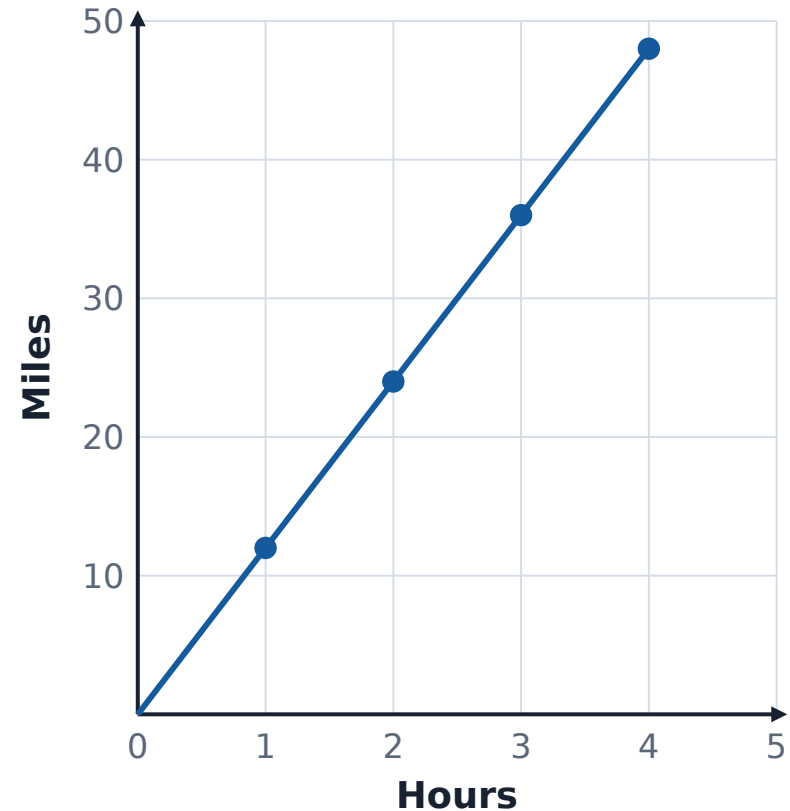
Walker A: $d = 3.5t$. Walker B: the graph passes through $(0, 0)$ and $(2, 8)$.

Who is faster?

- A.** Walker A
- B.** Walker B
- C.** They are the same.

What is the speed shown by the graph in miles per hour?

Speed: _____



True or false?

A line through $(0, 0)$ and $(5, 300)$ shows a greater speed than $d = 65t$.

A. True

B. False

Compare unit rates to find the better buy or the faster rate

MAIN	Grade 6 main question	6.RP.A.3.b
BACKWARD 1	Dividing money amounts	5.NBT.B.7
BACKWARD 2	Comparing decimals	5.NBT.A.3.b
BACKWARD 3	Finding the amount for one	3.OA.A.2
FORWARD 1	Compare deals using percents	7.RP.A.3
FORWARD 2	Compare two functions represented in different ways	8.F.A.2

SET 9 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

Which is the better buy?

A: 6 bottles of water for \$4.50

B: 10 bottles of water for \$7.00

A. A

B. B

C. Both are the same.

Which is the better buy?

A: a 12-ounce box of cereal for \$3.60

B: an 18-ounce box of cereal for \$4.95

A. A

B. B

C. Both are the same.

Which store has the lower price per pound?

A: 4 pounds of grapes for \$7.00

B: 3 pounds of grapes for \$5.40

A. A

B. B

C. Both are the same.

Which is the better buy?

A: 8 batteries for \$10.00

B: 12 batteries for \$14.40

A. A

B. B

C. Both are the same.

Which printer is faster?

A: 90 pages in 6 minutes

B: 140 pages in 10 minutes

A. A

B. B

C. Both are the same.

SET 9 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$4.50 \div 6$$

Quotient: _____

Find the quotient.

$$\$3.60 \div 12$$

Quotient: _____

$$\$7.00 \div 10 = ?$$

- A.** \$0.07
- B.** \$0.70
- C.** \$7.00
- D.** \$70.00

Find the quotient.

$$\$14.40 \div 12$$

Quotient: _____

True or false?

$$\$5.40 \div 3 = \$1.80$$

A. True

B. False

SET 9 • 6.RP.A.3.b

BACKWARD 2

Comparing decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Which amount is less?

A. \$0.70

B. \$0.75

True or false?

$$0.8 > 0.75$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

0.275 0.3

Symbol: _____

Which number is greatest?

A. 1.075

B. 1.705

C. 1.75

D. 1.8

True or false?

$$1.25 < 1.2$$

A. True

B. False

SET 9 • 6.RP.A.3.b

BACKWARD 3

Finding the amount for one • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

3 bags hold 21 oranges in all, the same number in each.

How many oranges are in 1 bag?

Oranges: _____

A worker packs 32 boxes in 4 hours, the same number each hour.

How many boxes per hour?

Boxes: _____

True or false?

9 muffins cost \$18 in all, the same price each, so 1 muffin costs \$2.

A. True

B. False

40 pages are read in 8 days, the same number each day.

How many pages per day?

A. 5

B. 32

C. 48

D. 320

6 packs hold 54 cards in all, the same number in each.

How many cards are in 1 pack?

Cards: _____

SET 9 • 6.RP.A.3.b

FORWARD 1

Compare deals using percents • Grade 7 • Ratios & Proportional Relationships
• 7.RP.A.3

A \$60 jacket is on sale. Which deal saves more money?

- A.** \$10 off
- B.** 20% off
- C.** They save the same.

Which costs less after the discount?

A: a \$45 game at 30% off

B: a \$40 game at 20% off

A. A

B. B

C. They cost the same.

Store A sells a \$80 bike at 15% off. Store B sells the same bike for \$70.

Which store has the lower price, and by how much?

Answer: _____

Which is the better deal on a \$120 phone?

- A.** 25% off
- B.** \$25 off
- C.** They are the same.

True or false?

A \$50 item at 10% off costs the same as a \$55 item at 20% off.

- A. True
- B. False

SET 9 • 6.RP.A.3.b

FORWARD 2

Compare two functions represented in different ways • Grade 8 • Functions •
8.F.A.2

Plan A costs $y = 12x$ dollars for x months. Plan B is shown in the table.

Which plan has the greater rate of change?

Months (x)	2	4	6
Cost (y)	26	52	78

A. Plan A

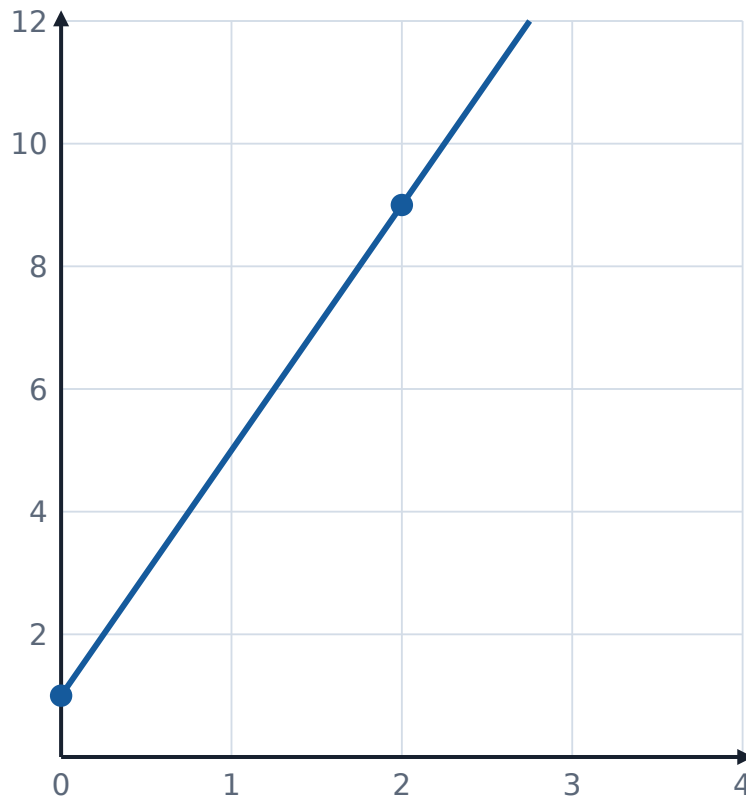
B. Plan B

C. They are the same.

Function A is $y = 3x + 4$. Function B is shown in the graph.

Which function has the greater rate of change?

- A.** Function A
- B.** Function B
- C.** They are the same.



Function A is shown in the table. Function B is $y = -3x + 2$.

Which function has the greater y-intercept?

x	0	1	2
y	5	3	1

- A.** Function A **B.** Function B **C.** They are the same.

Phone plan A costs \$30 plus \$0.05 per text. Phone plan B is a line through (0, 20) and (100, 35).

Which plan has the greater cost per text?

Plan: _____

True or false?

$y = 4x + 1$ has a greater rate of change than the function in the table.

x	0	1	2
y	0	5	10

A. True

B. False

Find a percent of a quantity as a rate per 100

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Hundredths as decimals	4.NF.C.6
BACKWARD 2	Tenths as hundredths	4.NF.C.5
BACKWARD 3	Fraction of a whole number	5.NF.B.4.a
BACKWARD 4	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Multi-step percent problems	7.RP.A.3
FORWARD 2	Build a linear function that uses a percent rate	8.F.B.4

SET 10 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 35% of 60?

Answer: _____

A jacket costs \$60. The sales tax is 8%.

How much is the tax?

Tax: _____

A class has 25 students. 40% of the students walk to school.

How many students walk to school?

Students: _____

A \$80 pair of shoes is 15% off.

How many dollars are taken off the price?

- A.** \$1.20
- B.** \$12
- C.** \$15
- D.** \$68

A school has 450 students. 12% of them play in the band.

How many students play in the band?

Students: _____

SET 10 • 6.RP.A.3.c

BACKWARD 1

Hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{45}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.07 = \frac{7}{100}$$

A. True

B. False

Which decimal is equal to $\frac{30}{100}$?

- A.** 0.03
- B.** 0.30
- C.** 3.0
- D.** 30.0

Write 0.58 as a fraction with a denominator of 100.

Fraction: _____

True or false?

$$\frac{8}{100} = 0.8$$

A. True

B. False

SET 10 • 6.RP.A.3.c

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Find the missing number.

$$\frac{3}{10} = \frac{?}{100}$$

Missing number: _____

True or false?

$$\frac{6}{10} + \frac{3}{100} = \frac{63}{100}$$

A. True

B. False

Which fraction is equal to $\frac{7}{10}$?

A. $\frac{7}{100}$

B. $\frac{10}{70}$

C. $\frac{17}{100}$

D. $\frac{70}{100}$

Add.

$$\frac{2}{10} + \frac{45}{100}$$

Sum: _____

True or false?

$$\frac{9}{10} = \frac{90}{100}$$

A. True

B. False

SET 10 • 6.RP.A.3.c

BACKWARD 3

Fraction of a whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{35}{100} \times 60$$

Product: _____

Find $\frac{1}{4}$ of 60.

Answer: _____

True or false?

$$\frac{2}{5} \times 25 = 10$$

A. True

B. False

$$\frac{3}{4} \times 20 = ?$$

- A.** 5
- B.** 12
- C.** 15
- D.** 16

Find $\frac{7}{10}$ of 50.

Answer: _____

SET 10 • 6.RP.A.3.c

BACKWARD 4

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$0.35 \times 60$$

Product: _____

Multiply.

$$0.08 \times 60$$

Product: _____

True or false?

$$0.12 \times 450 = 54$$

A. True

B. False

$$0.4 \times 25 = ?$$

A. 0.1

B. 1

C. 10

D. 100

Multiply.

$$0.15 \times 80$$

Product: _____

SET 10 • 6.RP.A.3.c

FORWARD 1

Multi-step percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A meal costs \$40. You leave a 15% tip.

What is the total cost of the meal with the tip?

Total: _____

A pair of shoes costs \$80. They are on sale for 25% off.
What is the sale price?

Sale price: _____

A town's population grew from 2,000 to 2,300.
What is the percent increase?

Percent increase: _____

A \$50 game is marked up 20%. Then a 5% sales tax is added to the new price.

What is the final cost?

A. \$60.00

B. \$62.00

C. \$62.50

D. \$63.00

A salesperson earns a 6% commission. She sells \$3,500 in products.

How much commission does she earn?

Commission: _____

SET 10 • 6.RP.A.3.c

FORWARD 2

Build a linear function that uses a percent rate • Grade 8 • Functions • 8.F.B.4

A salesperson earns \$300 per week plus 5% of her sales S .

Write a function for her weekly earnings E .

$E =$ _____

A server earns \$40 per shift plus 18% of the food sales f .

Which function gives the earnings E ?

A. $E = 0.18f - 40$

B. $E = 0.18f + 40$

C. $E = 18f + 40$

D. $E = 40f + 0.18$

An agent earns \$500 per month plus 3% of her sales r .
What are the rate of change and the initial value of this function?

Rate of change: _____

Initial value: _____

A delivery driver earns \$60 per day plus 10% of the orders d delivered in dollars.

Write a function for the daily pay P .

$P =$ _____

True or false?

For $E = 0.04s + 250$, earnings increase by \$0.04 for each \$1 of sales s .

- A. True
- B. False

Find the whole, given a part and the percent

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Benchmark percents as fractions	4.NF.A.1
BACKWARD 2	Finding an unknown factor	3.OA.A.4
BACKWARD 3	Dividing a whole number by a unit fraction	5.NF.B.7.b
FORWARD 1	Percent problems: find the original amount	7.RP.A.3
FORWARD 2	Linear equations with decimal coefficients	8.EE.C.7.b

SET 11 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

12 is 25% of what number?

Answer: _____

A student answered 18 questions correctly. This was 90% of the questions on the test.

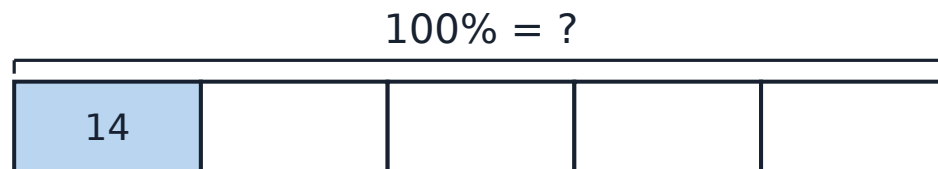
How many questions were on the test?

Questions: _____

Maria has saved \$45. This is 30% of the cost of a bike.
How much does the bike cost?

Cost: _____

The tape diagram shows that 20% of a number is 14.
What is the number?



A. 28

B. 34

C. 70

D. 280

In a survey, 36 students chose pizza. This was 40% of the students surveyed.

How many students were surveyed?

Students: _____

SET 11 • 6.RP.A.3.c

BACKWARD 1

Benchmark percents as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.A.1

Find the missing number.

$$\frac{25}{100} = \frac{1}{?}$$

Missing number: _____

Find the missing number.

$$\frac{50}{100} = \frac{?}{2}$$

Missing number: _____

True or false?

$$\frac{20}{100} = \frac{1}{5}$$

A. True

B. False

Which fraction is equivalent to $\frac{75}{100}$?

A. $\frac{1}{75}$

B. $\frac{7}{10}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

Find the missing number.

$$\frac{10}{100} = \frac{1}{?}$$

Missing number: _____

SET 11 • 6.RP.A.3.c

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$\square \times 4 = 36$$

$$\square = \underline{\hspace{10em}}$$

What number goes in the box?

$$5 \times \square = 45$$

$$\square = \underline{\hspace{10em}}$$

True or false?

If $6 \times \square = 42$, then $\square = 7$.

A. True

B. False

$$\square \times 8 = 56$$

What number goes in the box?

- A.** 6
- B.** 7
- C.** 8
- D.** 48

What number goes in the box?

$$9 \times \square = 63$$

$$\square = \underline{\hspace{10em}}$$

SET 11 • 6.RP.A.3.c

BACKWARD 3

Dividing a whole number by a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Divide.

$$12 \div \frac{1}{4}$$

Quotient: _____

Divide.

$$5 \div \frac{1}{3}$$

Quotient: _____

True or false?

$$6 \div \frac{1}{2} = 3$$

A. True

B. False

How many fifths are in 3?

A. $\frac{3}{5}$

B. 5

C. 8

D. 15

Divide.

$$8 \div \frac{1}{5}$$

Quotient: _____

SET 11 • 6.RP.A.3.c

FORWARD 1

Percent problems: find the original amount • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

After a 20% discount, a jacket costs \$48.

What was the original price?

Original price: _____

A price increased by 10%. The new price is \$55.
What was the original price?

Original price: _____

The sales tax on an item was \$3.60. The tax rate is 6%.
What was the price before tax?

Price: _____

A town's population grew by 25% to 1,500 people.
What was the population before the increase?

- A.** 1,125
- B.** 1,200
- C.** 1,475
- D.** 1,875

A store marks up the price of a bike by 40%. The selling price is \$210.

How much did the store pay for the bike?

Cost: _____

SET 11 • 6.RP.A.3.c

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$x - 0.2x = 48$$

x = _____

Solve for x.

$$0.4x + 12 = 0.1x + 30$$

x = _____

Solve for x.

$$0.25(x + 40) = 30$$

x = _____

Which value of n solves the equation?

$$n + 0.15n = 92$$

- A.** 78.2
- B.** 80
- C.** 92.15
- D.** 105.8

Solve for y .

$$2.5y - 7 = 1.5y + 3$$

$y =$ _____

Convert measurement units using ratio reasoning

MAIN	Grade 6 main question	6.RP.A.3.d
BACKWARD 1	Measurement equivalents	4.MD.A.1
BACKWARD 2	Multiplying and dividing by powers of 10	5.NBT.A.2
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Rates with fractions and unit conversions	7.RP.A.1
FORWARD 2	Convert units for very large and very small quantities	8.EE.A.4

SET 12 • 6.RP.A.3.d

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.d

There are 3 feet in 1 yard.

How many feet are in 14 yards?

Feet: _____

There are 12 inches in 1 foot.

How many inches are in 4.5 feet?

Inches: _____

There are 16 ounces in 1 pound.

How many pounds are in 56 ounces?

Pounds: _____

A recipe needs 3 quarts of milk. There are 4 cups in 1 quart.

How many cups of milk are needed?

A. $\frac{3}{4}$ cup

B. $1\frac{1}{3}$ cups

C. 7 cups

D. 12 cups

1 inch = 2.54 centimeters.

A book is 10 inches tall. How many centimeters tall is it?

Centimeters: _____

SET 12 • 6.RP.A.3.d

BACKWARD 1

Measurement equivalents • Grade 4 • Measurement & Data • 4.MD.A.1

How many ounces are in 1 pound?

Ounces: _____

True or false?

There are 12 inches in 1 yard.

A. True

B. False

How many cups are in 1 quart?

- A.** 2
- B.** 4
- C.** 8
- D.** 16

How many grams are in 1 kilogram?

Grams: _____

True or false?

1 meter = 100 centimeters

A. True

B. False

SET 12 • 6.RP.A.3.d

BACKWARD 2

Multiplying and dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$3.4 \times 1,000$$

Product: _____

Multiply.

$$2.54 \times 10$$

Product: _____

True or false?

$$0.6 \times 1,000 = 60$$

A. True

B. False

$$450 \div 100 = ?$$

- A.** 0.45
- B.** 4.5
- C.** 45
- D.** 45,000

Multiply.

$$7.25 \times 100$$

Product: _____

SET 12 • 6.RP.A.3.d

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$4.5 \times 12$$

Product: _____

Multiply.

$$3.5 \times 16$$

Product: _____

True or false?

$$1.5 \times 4 = 6$$

A. True

B. False

$$2.25 \times 4 = ?$$

A. 0.9

B. 8.1

C. 9

D. 90

Multiply.

$$6 \times 0.75$$

Product: _____

SET 12 • 6.RP.A.3.d

FORWARD 1

Rates with fractions and unit conversions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A faucet drips $\frac{1}{4}$ cup of water every $\frac{1}{3}$ minute.

How many cups does it drip per hour?

Cups per hour: _____

A snail travels $\frac{3}{8}$ foot every $\frac{1}{2}$ minute.

How many feet does it travel per hour?

Feet per hour: _____

A plant grows $\frac{1}{2}$ inch in $\frac{2}{3}$ week.

How many inches does it grow per week?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{3}{4}$

D. $1\frac{1}{3}$

A hose fills $\frac{3}{4}$ gallon in $\frac{1}{6}$ minute.

How many gallons does it fill per minute?

Gallons per minute: _____

A worker paints $\frac{2}{5}$ of a room in $\frac{1}{2}$ hour.

How many rooms can the worker paint in an 8-hour day?

Rooms: _____

SET 12 • 6.RP.A.3.d

FORWARD 2

Convert units for very large and very small quantities • Grade 8 • Expressions & Equations • 8.EE.A.4

$$1 \text{ km} = 10^6 \text{ mm.}$$

Write 3.2×10^6 millimeters in kilometers.

Kilometers: _____

Earth is about 1.5×10^8 km from the Sun. $1 \text{ km} = 10^3$ m.

How many meters is this? Write the answer in scientific notation.

Meters: _____

A sheet of paper is about 1×10^{-4} meter thick.

Which unit is the most appropriate for this measurement?

A. Miles

B. Meters

C. Millimeters

D. Kilometers

A virus is 1.2×10^{-7} meter long. 1 nanometer = 10^{-9} meter.

How many nanometers long is the virus?

Nanometers: _____

A glacier moves 3×10^{-1} meter per day. $1 \text{ m} = 10^2 \text{ cm}$.

How many centimeters does it move per day?

Centimeters: _____

Compute quotients of fractions and mixed numbers

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Multiplying fractions	5.NF.B.4.a
BACKWARD 2	Whole number $\div$ unit fraction	5.NF.B.7.b
BACKWARD 3	Unit fraction $\div$ whole number	5.NF.B.7.a
BACKWARD 4	Equivalent fractions	4.NF.A.1
BACKWARD 5	Mixed numbers as fractions	4.NF.B.3.b
FORWARD 1	Multiply and divide rational numbers	7.NS.A.2.c
FORWARD 2	Equations with fraction coefficients	8.EE.C.7.b

SET 13 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{5} \div \frac{2}{3}$$

Quotient: _____

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

Divide.

$$1\frac{1}{2} \div \frac{3}{4}$$

Quotient: _____

Which is equal to $\frac{4}{5} \div \frac{2}{3}$?

A. $\frac{2}{15}$

B. $\frac{8}{15}$

C. $\frac{5}{6}$

D. $1\frac{1}{5}$

Divide.

$$3\frac{1}{3} \div \frac{5}{6}$$

Quotient: _____

SET 13 • 6.NS.A.1

BACKWARD 1

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{5} \times \frac{2}{7}$$

Product: _____

True or false?

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

A. True

B. False

$$\frac{1}{2} \times \frac{5}{6} = ?$$

A. $\frac{1}{3}$

B. $\frac{5}{12}$

C. $\frac{5}{8}$

D. $\frac{6}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q3/5

Multiply.

$$\frac{4}{5} \times \frac{3}{2}$$

Product: _____

Multiply.

$$\frac{5}{6} \times 3$$

Product: _____

SET 13 • 6.NS.A.1

BACKWARD 2

Whole number $\div$ unit fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$3 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$4 \div \frac{1}{2} = 8$$

A. True

B. False

How many thirds are in 2?

A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. 5

D. 6

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

Divide.

$$2 \div \frac{1}{8}$$

Quotient: _____

SET 13 • 6.NS.A.1

BACKWARD 3

Unit fraction $\div$ whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{4} \div 3$$

Quotient: _____

True or false?

$$\frac{1}{2} \div 3 = \frac{1}{6}$$

A. True

B. False

$$\frac{1}{5} \div 2 = ?$$

A. $\frac{1}{10}$

B. $\frac{2}{5}$

C. $\frac{5}{2}$

D. 10

Divide.

$$\frac{1}{6} \div 3$$

Quotient: _____

True or false?

$$\frac{1}{6} \div 2 = \frac{2}{6}$$

A. True

B. False

SET 13 • 6.NS.A.1

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{8}{12}$ is equivalent to $\frac{2}{3}$.

A. True

B. False

Find the missing number.

$$\frac{10}{15} = \frac{?}{3}$$

Missing number: _____

Which fraction is equivalent to $\frac{9}{12}$?

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $\frac{12}{9}$

D. $\frac{9}{4}$

Find the missing number.

$$\frac{6}{8} = \frac{?}{4}$$

Missing number: _____

True or false?

$\frac{4}{6}$ is equivalent to $\frac{6}{8}$.

A. True

B. False

SET 13 • 6.NS.A.1

BACKWARD 5

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $1\frac{1}{2}$ as a fraction.

Fraction: _____

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$2\frac{1}{4} = \frac{9}{4}$$

A. True

B. False

Which fraction is equal to $1\frac{2}{3}$?

A. $\frac{3}{3}$

B. $\frac{5}{3}$

C. $\frac{7}{3}$

D. $\frac{12}{3}$

Write $\frac{11}{4}$ as a mixed number.

Mixed number: _____

SET 13 • 6.NS.A.1

FORWARD 1

Multiply and divide rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

Find the value.

$$\left(-\frac{2}{3}\right) \div \frac{4}{5}$$

Find the value.

$$-\frac{3}{4} \times \left(-\frac{8}{9}\right)$$

$$\left(-1\frac{1}{2}\right) \div \left(-\frac{3}{8}\right) = ?$$

A. -4

B. $-\frac{9}{16}$

C. $\frac{9}{16}$

D. 4

Find the value.

$$\frac{5}{6} \div \left(-\frac{5}{12}\right)$$

True or false?

$$\left(-\frac{1}{2}\right) \times \frac{4}{5} = -\frac{2}{5}$$

A. True

B. False

SET 13 • 6.NS.A.1

FORWARD 2

Equations with fraction coefficients • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{3}{4}(x - 2) = \frac{1}{2}x + 5$$

x = _____

Solve for x.

$$\frac{2}{3}x + 4 = \frac{1}{6}x + 7$$

x = _____

Solve for x.

$$\frac{x}{2} + \frac{x}{3} = 10$$

- A.** 5
- B.** 6
- C.** 12
- D.** 60

Solve for x.

$$\frac{1}{2}(4x + 6) = 3x - 5$$

x = _____

Solve for y.

$$\frac{5}{8}y - 1 = \frac{3}{8}y + 2$$

y = _____

Solve word problems involving division of fractions

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Division as "how many groups?"	3.OA.A.2
BACKWARD 2	Unit fractions build a fraction	3.NF.A.1
BACKWARD 3	Area with fractional side lengths	5.NF.B.4.b
BACKWARD 4	Word problems: dividing with unit fractions	5.NF.B.7.c
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Solve linear equations from contexts	8.EE.C.7.b

SET 14 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{2}{3}$ -cup scoops are in $\frac{5}{6}$ cup of flour?

Scoops: _____

A path is $\frac{3}{4}$ mile long. It is divided into sections that are each $\frac{1}{8}$ mile long.

How many sections are there?

Sections: _____

A rectangular patio has an area of $\frac{3}{5}$ square yard. Its length is $\frac{9}{10}$ yard.

What is its width?

Width: _____

A pitcher holds $\frac{9}{10}$ liter of juice. Each glass holds $\frac{3}{10}$ liter.

How many glasses can be filled?

A. $\frac{27}{100}$

B. $\frac{1}{3}$

C. 3

D. 6

Ana has $\frac{5}{6}$ pound of trail mix. She fills bags that each hold $\frac{5}{12}$ pound.

How many bags can she fill?

Bags: _____

SET 14 • 6.NS.A.1

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 4 are in 20?

Groups: _____

True or false?

$12 \div 3$ can mean "How many groups of 3 are in 12?"

- A. True
- B. False

Tom has 18 cookies. He puts 6 cookies in each bag.

Which equation finds the number of bags?

A. $18 \times 6 = \square$

B. $18 - 6 = \square$

C. $18 \div 6 = \square$

D. $6 \div 18 = \square$

24 students are split into teams of 8.

How many teams are there?

Teams: _____

How many 5s are in 35?

Answer: _____

SET 14 • 6.NS.A.1

BACKWARD 2

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

True or false?

$\frac{5}{6}$ is 5 parts of size $\frac{1}{6}$.

A. True

B. False

$\frac{3}{8}$ is made of how many parts of size $\frac{1}{8}$?

A. $\frac{1}{3}$

B. 3

C. 5

D. 8

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

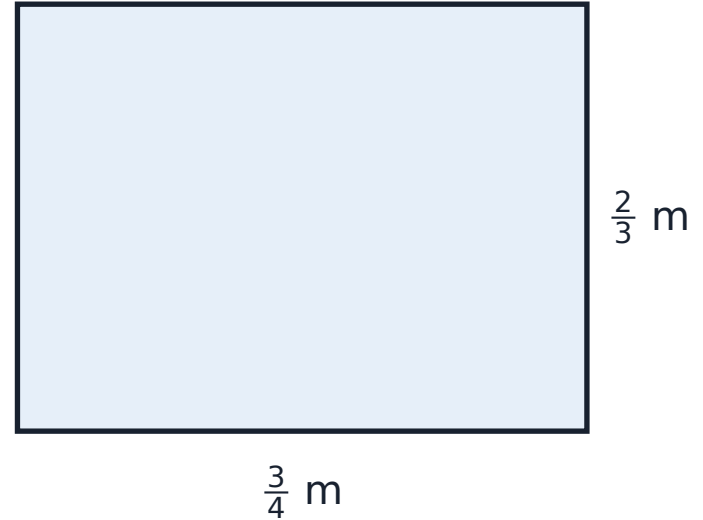
SET 14 • 6.NS.A.1

BACKWARD 3

Area with fractional side lengths • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

What is the area of the rectangle?

Area: _____



True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{3}$ foot has an area of $\frac{1}{6}$ square foot.

A. True

B. False

A rectangle is $\frac{2}{5}$ unit by $\frac{1}{2}$ unit.

What is its area?

A. $\frac{1}{5}$ square unit

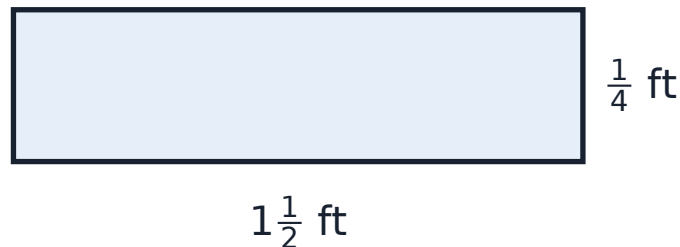
B. $\frac{2}{7}$ square unit

C. $\frac{3}{7}$ square unit

D. $\frac{9}{10}$ square unit

What is the area of the rectangle?

Area: _____



A garden is $\frac{5}{6}$ yard long and $\frac{3}{5}$ yard wide.

What is its area?

Area: _____

SET 14 • 6.NS.A.1

BACKWARD 4

Word problems: dividing with unit fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$ -pound burgers can be made from 3 pounds of beef?

Burgers: _____

A 4-pound bag of rice is divided into $\frac{1}{2}$ -pound portions.

How many portions are there?

Portions: _____

Which expression finds how many $\frac{1}{5}$ -mile laps are in 3 miles?

A. $\frac{1}{5} \div 3$

B. $3 - \frac{1}{5}$

C. $3 \div \frac{1}{5}$

D. $3 \times \frac{1}{5}$

4 friends share $\frac{1}{3}$ pound of cheese equally.

How much cheese does each friend get?

Each friend: _____

True or false?

There are 10 half-hours in 5 hours.

A. True

B. False

SET 14 • 6.NS.A.1

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot each second.

How many seconds does it take to descend $5\frac{1}{4}$ feet?

Seconds: _____

The temperature was 4°F . It dropped $2\frac{1}{2}^{\circ}\text{F}$ each hour for 3 hours.

What was the final temperature?

Temperature: _____

A stock's value changed by $-\$1\frac{1}{4}$ each day for 4 days.

What was the total change?

- A.** $-\$5$
- B.** $-\$4\frac{1}{4}$
- C.** $-\$1$
- D.** $\$5$

A recipe needs $\frac{2}{3}$ cup of oil per batch. Jo has $3\frac{1}{3}$ cups of oil.

How many batches can Jo make?

Batches: _____

A hiker descends 1,250 feet in $\frac{5}{6}$ hour at a constant rate.

What is the change in elevation per hour?

Change per hour: _____

SET 14 • 6.NS.A.1

FORWARD 2

Solve linear equations from contexts • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{1}{2}(x + 6) = \frac{1}{3}(x + 12)$$

x = _____

Lee's age plus $\frac{1}{3}$ of his age is 24.

Write and solve an equation to find Lee's age.

Age: _____

Tank A has 40 gallons and drains $\frac{3}{4}$ gallon per minute.

Tank B has 10 gallons and fills at $1\frac{1}{4}$ gallons per minute.

After how many minutes will the tanks hold the same amount?

Minutes: _____

Solve for x.

$$\frac{2}{5}x - 3 = \frac{1}{5}x + 1$$

A. -10

B. 4

C. 10

D. 20

Solve for n.

$$\frac{3}{4}n + \frac{1}{2} = \frac{1}{4}n + 4$$

n = _____

Divide multi-digit numbers using the standard algorithm

MAIN	Grade 6 main question	6.NS.B.2
BACKWARD 1	Multiplying multi-digit numbers	5.NBT.B.5
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Estimating quotients	5.NBT.B.6
BACKWARD 4	Multi-digit subtraction	4.NBT.B.4
FORWARD 1	Long division: fractions to decimals	7.NS.A.2.d
FORWARD 2	Repeating decimals and rational numbers	8.NS.A.1

SET 15 • 6.NS.B.2

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$4,508 \div 23$$

Quotient: _____

Use the standard algorithm to divide.

$$7,236 \div 36$$

Quotient: _____

Use the standard algorithm to divide.

$$9,975 \div 75$$

Quotient: _____

Use the standard algorithm to divide.

$$1,680 \div 48$$

A. 30

B. 35

C. 36

D. 350

Use the standard algorithm to divide.

$$25,494 \div 42$$

Quotient: _____

SET 15 • 6.NS.B.2

BACKWARD 1

Multiplying multi-digit numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$46 \times 23$$

Product: _____

Multiply.

$$196 \times 23$$

Product: _____

True or false?

$$35 \times 48 = 1,680$$

A. True

B. False

$$133 \times 75 = ?$$

- A.** 1,596
- B.** 9,775
- C.** 9,875
- D.** 9,975

Multiply.

$$607 \times 42$$

Product: _____

SET 15 • 6.NS.B.2

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 4$$

Quotient: _____

Divide.

$$1,256 \div 8$$

Quotient: _____

True or false?

$$936 \div 3 = 312$$

A. True

B. False

$$714 \div 7 = ?$$

- A.** 12
- B.** 102
- C.** 120
- D.** 1,002

Divide.

$$2,345 \div 5$$

Quotient: _____

SET 15 • 6.NS.B.2

BACKWARD 3

Estimating quotients • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Which is the best estimate for $4,508 \div 23$?

- A.** 2
- B.** 20
- C.** 200
- D.** 2,000

Which is the best estimate for $7,236 \div 36$?

- A.** 2
- B.** 20
- C.** 200
- D.** 2,000

True or false?

9,975 $\div$ 75 is between 100 and 200.

A. True

B. False

Estimate $6,120 \div 29$ by rounding 29 to 30.

Estimate: _____

How many digits are in the quotient of $1,680 \div 48$?

A. 1

B. 2

C. 3

D. 4

SET 15 • 6.NS.B.2

BACKWARD 4

Multi-digit subtraction • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Subtract.

$$5,012 - 4,698$$

Difference: _____

Subtract.

$$3,000 - 1,847$$

Difference: _____

True or false?

$$7,236 - 7,200 = 36$$

A. True

B. False

$$9,975 - 9,000 = ?$$

- A.** 75
- B.** 975
- C.** 1,975
- D.** 9,075

Subtract.

$$6,203 - 2,875$$

Difference: _____

SET 15 • 6.NS.B.2

FORWARD 1

Long division: fractions to decimals • Grade 7 • The Number System •
7.NS.A.2.d

Use long division to write $\frac{5}{11}$ as a decimal.

Decimal: _____

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

Which fraction has a repeating decimal?

A. $\frac{1}{5}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{7}{8}$

True or false?

The decimal form of $\frac{7}{20}$ terminates.

A. True

B. False

Use long division to write $\frac{5}{6}$ as a decimal.

Decimal: _____

SET 15 • 6.NS.B.2

FORWARD 2

Repeating decimals and rational numbers • Grade 8 • The Number System •
8.NS.A.1

Write $0.272727\ldots$ as a fraction in simplest form.

Fraction: _____

Write 0.444... as a fraction.

Fraction: _____

Which number is irrational?

A. 0.333...

B. $\frac{5}{8}$

C. $\sqrt{16}$

D. $\sqrt{7}$

True or false?

Every repeating decimal is a rational number.

A. True

B. False

Write 1.1666... as a fraction.

Fraction: _____

Add multi-digit decimals

MAIN	Grade 6 main question
BACKWARD 1	Place value of decimal digits
BACKWARD 2	Equivalent decimals
BACKWARD 3	Lining up decimal points to add
BACKWARD 4	Multi-digit addition
FORWARD 1	Add rational numbers
FORWARD 2	Linear equations with decimal coefficients

6.NS.B.3

5.NBT.A.3.a

5.NBT.A.3.b

5.NBT.B.7

4.NBT.B.4

7.NS.A.1.d

8.EE.C.7.b

SET 16 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the sum.

$$315.4 + 28.75 + 6.309$$

Sum: _____

Find the sum.

$$47.08 + 9.6$$

Sum: _____

Find the sum.

$$128.5 + 3.47 + 60.009$$

Sum: _____

Find the sum.

$$6.75 + 12.9 + 0.48$$

- A.** 7.23
- B.** 19.23
- C.** 20.03
- D.** 20.13

Find the sum.

$$0.875 + 14.2 + 3.06$$

Sum: _____

SET 16 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.358, what place value does the digit 5 have?

- A.** Tenths
- B.** Hundredths
- C.** Ones
- D.** Thousandths

In 42.906, which digit is in the tenths place?

Digit: _____

True or false?

In 3.47, the digit 7 is in the hundredths place.

A. True

B. False

What is the value of the 8 in 0.584?

A. 0.008

B. 0.08

C. 0.8

D. 8

Write $5 + 0.3 + 0.07$ as a decimal.

Decimal: _____

SET 16 • 6.NS.B.3

BACKWARD 2

Equivalent decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

True or false?

2.4 and 2.40 have the same value.

A. True

B. False

True or false?

5.6 and 5.06 have the same value.

A. True

B. False

Which decimal is equal to 7.5?

- A.** 0.75
- B.** 7.005
- C.** 7.05
- D.** 7.50

Write 9.6 with a zero in the hundredths place.

Decimal: _____

True or false?

$$0.300 = 0.3$$

A. True

B. False

SET 16 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to add • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $36.2 + 4.58$?

A.

$$\begin{array}{r} 36.2 \\ +4.58 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 36.20 \\ + 4.58 \\ \hline \end{array}$$

Which setup lines up the decimal points for $5.07 + 12.3$?

A.

$$\begin{array}{r} 5.07 \\ +12.30 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 5.07 \\ +12.3 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

A. True

B. False

$$\begin{array}{r} 8.4 \\ +2.75 \\ \hline \end{array}$$

Which setup lines up the decimal points for $128.5 + 3.47$?

A.

$$\begin{array}{r} 128.50 \\ + 3.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 128.5 \\ + 3.47 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 9.30 \\ + 0.45 \\ \hline \end{array}$$

SET 16 • 6.NS.B.3

BACKWARD 4

Multi-digit addition • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$3,154 + 2,875$$

Sum: _____

Add.

$$4,605 + 2,798$$

Sum: _____

True or false?

$$1,975 + 3,489 = 5,464$$

A. True

B. False

$$6,758 + 1,265 = ?$$

A. 7,013

B. 7,923

C. 8,013

D. 8,023

Add.

$$12,480 + 9,635$$

Sum: _____

SET 16 • 6.NS.B.3

FORWARD 1

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-18.6 + 4.35 =$$

Find the value.

$$-24.5 + 3.27 =$$

Find the value.

$$-3.4 + (-9.15) =$$

Find the value.

$$-6.25 + (-2.8) = ?$$

A. -9.05

B. -3.45

C. 3.45

D. 9.05

Find the value.

$$-15.75 + 20.3 =$$

SET 16 • 6.NS.B.3

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$-2.4x + 7.8 = 1.2x - 1.2$$

x = _____

Solve for x.

$$-1.5x + 6.4 = 0.5x - 1.6$$

x = _____

Solve for x.

$$0.8x + 7.2 = -1.2x + 1.2$$

x = _____

Solve for x.

$$-4.5x - 3.25 = -2.5x + 0.75$$

x = _____

Solve for x.

$$2.4(x + 1.5) = 1.4x + 6.6$$

x = _____

Subtract multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Place value of decimal digits

5.NBT.A.3.a

BACKWARD 2

Writing zeros to make equal decimal places

5.NBT.A.3.b

BACKWARD 3

Lining up decimal points to subtract

5.NBT.B.7

BACKWARD 4

Subtracting across zeros

4.NBT.B.4

FORWARD 1

Subtract rational numbers by adding the additive inverse

7.NS.A.1.c

FORWARD 2

Solve equations by collecting like terms

8.EE.C.7.b

SET 17 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the difference.

$$52.3 - 18.47$$

Difference: _____

Find the difference.

$$100 - 36.58$$

Difference: _____

Find the difference.

$$7.4 - 2.068$$

Difference: _____

Find the difference.

$$15.02 - 6.9$$

- A.** 8.12
- B.** 8.93
- C.** 9.12
- D.** 14.33

Find the difference.

$$243.6 - 87.95$$

Difference: _____

SET 17 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 2.068, what place value does the digit 6 have?

- A.** Tenths
- B.** Thousandths
- C.** Hundredths
- D.** Ones

In 87.95, which digit is in the hundredths place?

Digit: _____

True or false?

In 36.58, the digit 5 is in the tenths place.

- A. True
- B. False

What is the value of the 4 in 7.04?

- A.** 0.004
- B.** 0.04
- C.** 0.4
- D.** 4

Write $20 + 0.6 + 0.009$ as a decimal.

Decimal: _____

SET 17 • 6.NS.B.3

BACKWARD 2

Writing zeros to make equal decimal places • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.3.b

True or false?

$$7.4 = 7.400$$

A. True

B. False

Write 100 with two decimal places.

Decimal: _____

Which decimal is equal to 15.1?

- A.** 1.51
- B.** 15.001
- C.** 15.01
- D.** 15.10

Write 52.3 with a digit in the hundredths place, without changing its value.

Decimal: _____

True or false?

$$6.9 = 6.09$$

A. True

B. False

SET 17 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to subtract • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Which setup lines up the decimal points for $15.6 - 3.42$?

A.

$$\begin{array}{r} 15.60 \\ - 3.42 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 15.6 \\ - 3.42 \\ \hline \end{array}$$

Which setup lines up the decimal points for $52.3 - 18.47$?

A.

$$\begin{array}{r} 52.3 \\ -18.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 52.30 \\ -18.47 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A. True
- B. False

$$\begin{array}{r} 7.4 \\ -2.068 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

A. True

B. False

$$\begin{array}{r} 100.00 \\ - 36.58 \\ \hline \end{array}$$

Which setup lines up the decimal points for $243.6 - 87.95$?

A.

$$\begin{array}{r} 243.60 \\ - 87.95 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 243.6 \\ - 87.95 \\ \hline \end{array}$$

SET 17 • 6.NS.B.3

BACKWARD 4

Subtracting across zeros • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Subtract.

$$5,000 - 2,847$$

Difference: _____

Subtract.

$$3,006 - 1,478$$

Difference: _____

True or false?

$$6,000 - 2,458 = 3,542$$

A. True

B. False

$$8,014 - 3,276 = ?$$

A. 4,738

B. 4,748

C. 4,838

D. 5,262

Subtract.

$$10,000 - 3,658$$

Difference: _____

SET 17 • 6.NS.B.3

FORWARD 1

Subtract rational numbers by adding the additive inverse • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$-4.2 - 3.75$$

Value: _____

Find the value.

$$2.5 - (-6.8)$$

Value: _____

Which expression is equal to $8.1 - 12.4$?

- A.** $12.4 - 8.1$
- B.** $-8.1 + 12.4$
- C.** $8.1 + (-12.4)$
- D.** $8.1 + 12.4$

Find the value.

$$-0.65 - (-1.2)$$

Value: _____

True or false?

$$-7.5 - 2.5 = -7.5 + (-2.5)$$

A. True

B. False

SET 17 • 6.NS.B.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$5.2x - 3.75 = 2.2x + 8.25$$

x = _____

Solve for x.

$$0.9x + 12.5 = 0.4x + 10$$

x = _____

Solve for x.

$$6.3 - 1.2x = 0.8x - 1.7$$

A. -4

B. 0.4

C. 2.5

D. 4

Solve for x.

$$4.75 - x = 2x - 1.25$$

x = _____

Solve for x.

$$3.4x - 9.6 = 1.4x - 2.6$$

x = _____

Find and correct decimal alignment errors in addition and subtraction

MAIN	Grade 6 main question	6.NS.B.3
BACKWARD 1	Lining up decimal points	5.NBT.B.7
BACKWARD 2	Place value of decimal digits	5.NBT.A.3.a
BACKWARD 3	Estimating to check reasonableness	5.NBT.A.4
FORWARD 1	Find errors when adding and subtracting rational numbers	7.NS.A.1.d
FORWARD 2	Find errors in solving linear equations	8.EE.C.7.b

SET 18 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Jon put this example on the board.

What mistake was made?

What should the answer be?

$$\begin{array}{r} 315.2 \\ 24.8 \\ +7.936 \\ \hline 11.336 \end{array}$$

Mistake: _____

Correct answer: _____

Ava put this example on the board.

What mistake was made?

What should the answer be?

$$\begin{array}{r} 56.4 \\ +3.25 \\ \hline 8.89 \end{array}$$

Mistake: _____

Correct answer: _____

Leo put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 48.6 \\ -2.15 \\ \hline 2.71 \end{array}$$

Mistake: _____

Correct answer: _____

Mia put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline 8.34 \end{array}$$

Mistake: _____

Correct answer: _____

Ben put this example on the board.

What mistake was made?

What should the answer be?

$$\begin{array}{r} 125.6 \\ -3.48 \\ \hline 9.08 \end{array}$$

Mistake: _____

Correct answer: _____

SET 18 • 6.NS.B.3

BACKWARD 1

Lining up decimal points • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $56.4 + 3.25$?

A.

$$\begin{array}{r} 56.4 \\ + 3.25 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 56.40 \\ + 3.25 \\ \hline \end{array}$$

Which setup lines up the decimal points for $48.6 - 2.15$?

A.

$$\begin{array}{r} 48.60 \\ - 2.15 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 48.6 \\ - 2.15 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A. True
- B. False

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline \end{array}$$

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 125.60 \\ - 3.48 \\ \hline \end{array}$$

Which setup lines up the decimal points for $24.8 + 7.936$?

A.

$$\begin{array}{r} 24.8 \\ + 7.936 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 24.800 \\ + 7.936 \\ \hline \end{array}$$

SET 18 • 6.NS.B.3

BACKWARD 2

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.936, what place value does the digit 3 have?

- A.** Hundredths
- B.** Tenths
- C.** Thousandths
- D.** Ones

In 315.2, which digit is in the ones place?

Digit: _____

True or false?

In 3.25, the digit 2 is in the tenths place.

A. True

B. False

What is the value of the 6 in 12.56?

- A.** 0.06
- B.** 0.6
- C.** 6
- D.** 60

In 48.6, which digit is in the tenths place?

Digit: _____

SET 18 • 6.NS.B.3

BACKWARD 3

Estimating to check reasonableness • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round each number to the nearest whole number. Then add to estimate.

$$315.2 + 24.8 + 7.936$$

Estimate: _____

Round 48.6 to the nearest whole number.

Rounded: _____

True or false?

7.08 + 12.6 is about 20.

A. True

B. False

Which is the best estimate for $125.6 - 3.48$?

- A.** 9
- B.** 90
- C.** 122
- D.** 130

Round 56.4 and 3.25 to the nearest whole numbers.
Then add.

Estimate: _____

SET 18 • 6.NS.B.3

FORWARD 1

Find errors when adding and subtracting rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Ray wrote: $-3.5 + 1.25 = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Tia wrote: $-6.2 - 2.4 = -3.8$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Sam wrote: $4.5 - (-1.5) = 3$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Kai wrote: $-7.25 + (-2.5) = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

Lia wrote: $-0.8 + 2.3 = -1.5$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

SET 18 • 6.NS.B.3

FORWARD 2

Find errors in solving linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Kim solved $2.4x + 1.2 = 0.4x - 3.8$.

Step 1: $2x + 1.2 = -3.8$

Step 2: $2x = -2.6$

Step 3: $x = -1.3$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Jo solved $3.5x - 2 = 1.5x + 6$.

Step 1: $2x - 2 = 6$

Step 2: $2x = 4$

Step 3: $x = 2$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Ali solved $-1.2x + 5 = 0.8x - 3$.

Step 1: $-2x + 5 = -3$

Step 2: $-2x = -8$

Step 3: $x = -4$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Ned solved $0.5(x + 6) = 2x - 3$.

Step 1: $0.5x + 6 = 2x - 3$

Step 2: $9 = 1.5x$

Step 3: $x = 6$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Eve solved $4.2x - 1.5 = 2.2x + 2.5$.

Step 1: $2x - 1.5 = 2.5$

Step 2: $2x = 1$

Step 3: $x = 0.5$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

Multiply multi-digit decimals

- MAIN** Grade 6 main question
- BACKWARD 1** Multiplying whole numbers
- BACKWARD 2** Powers of 10 and the decimal point
- BACKWARD 3** Multiplying tenths and hundredths
- BACKWARD 4** Rounding decimals to estimate
- FORWARD 1** Multiply rational numbers
- FORWARD 2** Multiply numbers in scientific notation

6.NS.B.3

5.NBT.B.5

5.NBT.A.2

5.NBT.B.7

5.NBT.A.4

7.NS.A.2.c

8.EE.A.4

SET 19 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.6 \times 0.45$$

Product: _____

Multiply.

$$12.5 \times 3.8$$

Product: _____

Multiply.

$$4.25 \times 2.4$$

Product: _____

Multiply.

$$0.07 \times 0.6$$

A. 0.0042

B. 0.042

C. 0.42

D. 4.2

Multiply.

$$18.4 \times 0.35$$

Product: _____

SET 19 • 6.NS.B.3

BACKWARD 1

Multiplying whole numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$36 \times 45$$

Product: _____

Multiply.

$$125 \times 38$$

Product: _____

True or false?

$$425 \times 24 = 10,200$$

A. True

B. False

$$184 \times 35 = ?$$

- A.** 1,472
- B.** 5,440
- C.** 6,340
- D.** 6,440

Multiply.

$$208 \times 15$$

Product: _____

SET 19 • 6.NS.B.3

BACKWARD 2

Powers of 10 and the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$0.45 \times 10$$

Product: _____

Multiply.

$$3.6 \times 100$$

Product: _____

True or false?

$$24 \div 10 = 2.4$$

A. True

B. False

$$1,620 \div 1,000 = ?$$

- A.** 0.162
- B.** 1.62
- C.** 16.2
- D.** 162

Divide.

$$475 \div 10$$

Quotient: _____

SET 19 • 6.NS.B.3

BACKWARD 3

Multiplying tenths and hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$0.3 \times 0.2$$

Product: _____

Multiply.

$$0.5 \times 0.8$$

Product: _____

True or false?

$$0.4 \times 0.4 = 1.6$$

A. True

B. False

$$0.2 \times 0.03 = ?$$

A. 0.006

B. 0.06

C. 0.6

D. 6

Multiply.

$$1.2 \times 0.3$$

Product: _____

SET 19 • 6.NS.B.3

BACKWARD 4

Rounding decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.62 to the nearest whole number.

Rounded: _____

Round 12.48 to the nearest tenth.

Rounded: _____

True or false?

4.25 rounded to the nearest whole number is 4.

A. True

B. False

Round 0.456 to the nearest hundredth.

A. 0.4

B. 0.45

C. 0.46

D. 0.5

Round 8.75 to the nearest whole number.

Rounded: _____

SET 19 • 6.NS.B.3

FORWARD 1

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the value.

$$(-3.6)(0.45)$$

Find the value.

$$(-2.5)(-4.2)$$

$$-0.8 \times 1.5 = ?$$

- A.** -1.2
- B.** -0.12
- C.** 0.12
- D.** 1.2

Find the value.

$$(-1.2)(-0.5)(-3)$$

True or false?

$$(-0.6)(0.7) = 0.42$$

A. True

B. False

SET 19 • 6.NS.B.3

FORWARD 2

Multiply numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Multiply. Write the answer in scientific notation.

$$(3.6 \times 10^4)(4.5 \times 10^{-2})$$

Product: _____

Multiply. Write the answer in scientific notation.

$$(2.5 \times 10^3)(4 \times 10^5)$$

Product: _____

$$(1.2 \times 10^6)(3 \times 10^{-4}) = ?$$

- A.** 4.2×10^2
- B.** 3.6×10^{10}
- C.** 3.6×10^2
- D.** 3.6×10^{-2}

A light-year is about 9.5×10^{12} km.

About how many kilometers are in 4.2 light-years? Write the answer in scientific notation.

Kilometers: _____

True or false?

$$(5 \times 10^3)(6 \times 10^2) = 3 \times 10^6$$

A. True

B. False

Divide multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Dividing by a two-digit number

5.NBT.B.6

BACKWARD 2

Using powers of 10 to shift the decimal point

5.NBT.A.2

BACKWARD 3

Dividing tenths and hundredths

5.NBT.B.7

FORWARD 1

Divide rational numbers

7.NS.A.2.b

FORWARD 2

Divide numbers in scientific notation

8.EE.A.4

SET 20 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$15.75 \div 0.35$$

Quotient: _____

Divide.

$$8.64 \div 2.4$$

Quotient: _____

Divide.

$$12.6 \div 0.45$$

Quotient: _____

Divide.

$$0.72 \div 0.08$$

A. 0.09

B. 0.9

C. 9

D. 90

Divide.

$$283.5 \div 13.5$$

Quotient: _____

SET 20 • 6.NS.B.3

BACKWARD 1

Dividing by a two-digit number • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,575 \div 35$$

Quotient: _____

Divide.

$$864 \div 24$$

Quotient: _____

True or false?

$$1,260 \div 45 = 28$$

A. True

B. False

$$672 \div 32 = ?$$

- A.** 12
- B.** 21
- C.** 22
- D.** 210

Divide.

$$2,835 \div 15$$

Quotient: _____

SET 20 • 6.NS.B.3

BACKWARD 2

Using powers of 10 to shift the decimal point • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.2

Which has the same quotient as $15.75 \div 0.35$?

A. $1,575 \div 35$

B. $157.5 \div 35$

C. $1,575 \div 3.5$

D. $15.75 \div 35$

True or false?

$8.64 \div 2.4$ has the same quotient as $86.4 \div 24$.

A. True

B. False

Multiply.

$$0.45 \times 100$$

Product: _____

Multiply.

$$12.6 \times 100$$

Product: _____

$$0.08 \times \underline{\hspace{1cm}} = 8$$

- A.** 0.1
- B.** 10
- C.** 100
- D.** 1,000

SET 20 • 6.NS.B.3

BACKWARD 3

Dividing tenths and hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Divide.

$$2.4 \div 0.6$$

Quotient: _____

Divide.

$$3.5 \div 0.5$$

Quotient: _____

True or false?

$$1.2 \div 0.3 = 0.4$$

A. True

B. False

$$0.9 \div 0.3 = ?$$

A. 0.03

B. 0.3

C. 3

D. 30

Divide.

$$4.8 \div 4$$

Quotient: _____

SET 20 • 6.NS.B.3

FORWARD 1

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the value.

$$(-15.75) \div 0.35$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q1/5

$$-8.64 \div (-2.4)$$

ID S20-F1-Q2

785

True or false?

$$-(12 \div 4) = (-12) \div 4$$

A. True

B. False

$$7.2 \div (-0.9) = ?$$

A. -8

B. -0.8

C. 0.8

D. 8

Find the value.

$$-0.72 \div 0.08$$

SET 20 • 6.NS.B.3

FORWARD 2

Divide numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Divide. Write the answer in scientific notation.

$$(6.3 \times 10^8) \div (2.1 \times 10^3)$$

Quotient: _____

Divide. Write the answer in scientific notation.

$$(4.8 \times 10^{-2}) \div (1.6 \times 10^{-5})$$

Quotient: _____

$$(9 \times 10^6) \div (3 \times 10^9) = ?$$

A. 6×10^{-3}

B. 3×10^{15}

C. 3×10^3

D. 3×10^{-3}

Divide. Write the answer in scientific notation.

$$(7.5 \times 10^4) \div (2.5 \times 10^{-1})$$

Quotient: _____

True or false?

$$(8.4 \times 10^5) \div (4.2 \times 10^2) = 2 \times 10^3$$

A. True

B. False

Find the greatest common factor of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Factors	4.OA.B.4
BACKWARD 2	Prime and composite numbers	4.OA.B.4
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Factor linear expressions using a common factor	7.EE.A.1
FORWARD 2	Solve equations using the distributive property	8.EE.C.7.b

SET 21 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

What is the greatest common factor of 18 and 45?

- A.** 3
- B.** 9
- C.** 15
- D.** 90

Find the greatest common factor of 28 and 42.

GCF: _____

Find the greatest common factor of 30 and 75.

GCF: _____